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Method for disc therapy

Abstract

A method is provided that includes providing an electrode, which includes a wire that has a wire diameter of between 75 and 125 microns. The wire includes a non-electrically-insulated current-application longitudinal segment, which, in the absence of any applied forces, is coiled and has (i) an outer coil diameter of between 3 and 7 times the wire diameter, and (ii) an entire longitudinal length of between 5 and 35 mm. The wire further includes an electrically-insulated lead longitudinal segment, which has an entire longitudinal length of at least 10 mm, in the absence of any applied forces. At least a portion of the electrode is implanted in a body of a subject. Other embodiments are also described.

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References Cited

U.S. PATENT DOCUMENTS

Patent No.	Issued Date	Patentee Name	U.S. Cl.	CPC
2959410	12/1959	Fullam et al.	N/A	N/A
4044774	12/1976	Corbin et al.	N/A	N/A
4503863	12/1984	Katims	N/A	N/A
5088977	12/1991	Sibalis	N/A	N/A
5121754	12/1991	Mullett	N/A	N/A
5433739	12/1994	Sluijter et al.	N/A	N/A
5529574	12/1995	Frackelton	N/A	N/A
5587297	12/1995	Jacobson et al.	N/A	N/A
5792100	12/1997	Shantha	N/A	N/A
5911223	12/1998	Weaver et al.	N/A	N/A
5919163	12/1998	Glickman	N/A	N/A
5938690	12/1998	Law et al.	N/A	N/A
6041252	12/1999	Walker et al.	N/A	N/A
6146380	12/1999	Racz et al.	N/A	N/A
6161047	12/1999	King et al.	N/A	N/A
6360750	12/2001	Gerber et al.	N/A	N/A
6416510	12/2001	Altman	606/41	A61B 18/1492
6506408	12/2002	Palasis	N/A	N/A
6567702	12/2002	Nekhendzy et al.	N/A	N/A
6591138	12/2002	Fischell et al.	N/A	N/A
6602248	12/2002	Sharps et al.	N/A	N/A
6620155	12/2002	Underwood et al.	N/A	N/A
6907295	12/2004	Gross et al.	N/A	N/A
6941172	12/2004	Nachum	N/A	N/A
6997941	12/2005	Sharkey et al.	N/A	N/A
7120489	12/2005	Shalev et al.	N/A	N/A
7217351	12/2006	Krumme	N/A	N/A
7223227	12/2006	Pflueger	N/A	N/A
7270659	12/2006	Ricart et al.	N/A	N/A
7398121	12/2007	Matsumura et al.	N/A	N/A
7509171	12/2008	DiMauro	N/A	N/A

7640062	12/2008	Shalev	N/A	N/A
7831306	12/2009	Finch et al.	N/A	N/A
7860569	12/2009	Solberg et al.	N/A	N/A
8190248	12/2011	Besio et al.	N/A	N/A
8457761	12/2012	Wariar	N/A	N/A
8577469	12/2012	Gross	N/A	N/A
8676348	12/2013	Gross	N/A	N/A
9616221	12/2016	Gross	N/A	N/A
9724513	12/2016	Lane et al.	N/A	N/A
9770591	12/2016	Gross et al.	N/A	N/A
9950156	12/2017	Gross et al.	N/A	N/A
2002/0151948	12/2001	King et al.	N/A	N/A
2002/0183683	12/2001	Lerner	N/A	N/A
2003/0130707	12/2002	Gan et al.	N/A	N/A
2003/0158589	12/2002	Katsnelson	N/A	N/A
2003/0216792	12/2002	Levin et al.	N/A	N/A
2003/0225331	12/2002	Diederich et al.	N/A	N/A
2004/0002746	12/2003	Ryan et al.	N/A	N/A
2004/0019381	12/2003	Pflueger	N/A	N/A
2004/0049180	12/2003	Sharps et al.	N/A	N/A
2004/0116977	12/2003	Finch et al.	N/A	N/A
2004/0210209	12/2003	Yeung et al.	N/A	N/A
2004/0225363	12/2003	Richelsoph	N/A	N/A
2004/0253304	12/2003	Gross et al.	N/A	N/A
2004/0267240	12/2003	Gross et al.	N/A	N/A
2005/0010205	12/2004	Hovda et al.	N/A	N/A
2005/0021104	12/2004	DiLorenzo	N/A	N/A
2005/0058701	12/2004	Gross et al.	N/A	N/A
2005/0119650	12/2004	Sanders et al.	N/A	N/A
2005/0159790	12/2004	Shalev	N/A	N/A
2005/0277996	12/2004	Podhajsky et al.	N/A	N/A
2006/0030895	12/2005	Simon et al.	N/A	N/A
2006/0106430	12/2005	Fowler et al.	N/A	N/A
2006/0224223	12/2005	Podhajsky et al.	N/A	N/A
2006/0276844	12/2005	Alon et al.	N/A	N/A
2006/0293723	12/2005	Whitehurst et al.	N/A	N/A
2007/0000784	12/2006	Paul et al.	N/A	N/A
2007/0073402	12/2006	Vresilovic et al.	N/A	N/A
2007/0213700	12/2006	Davison et al.	N/A	N/A
2008/0009927	12/2007	Vilims	N/A	N/A
2008/0063703	12/2007	Gross et al.	N/A	N/A
2008/0119907	12/2007	Stahmann	N/A	N/A
2008/0188837	12/2007	Belsky et al.	N/A	N/A
2008/0260542	12/2007	Nishikawa et al.	N/A	N/A
2008/0275430	12/2007	Belsky et al.	N/A	N/A
2009/0112278	12/2008	Wingeier et al.	N/A	N/A
2009/0125080	12/2008	Montgomery	N/A	N/A
2009/0126813	12/2008	Yanagisawa et al.	N/A	N/A
2009/0131850	12/2008	Geiger	N/A	N/A
2009/0312816	12/2008	Gross	N/A	N/A

2010/0217369	12/2009	Gross	N/A	N/A
2010/0324441	12/2009	Hargrove et al.	N/A	N/A
2011/0046540	12/2010	Alterman et al.	N/A	N/A
2011/0160638	12/2010	Mauge et al.	N/A	N/A
2011/0160797	12/2010	Makous et al.	N/A	N/A
2012/0203307	12/2011	Schroeppel et al.	N/A	N/A
2013/0066392	12/2012	Simon et al.	N/A	N/A
2013/0102952	12/2012	Gross	N/A	N/A
2013/0166006	12/2012	Williams	N/A	N/A
2014/0058189	12/2013	Stubbeman	N/A	N/A
2014/0088672	12/2013	Bedenbaugh	N/A	N/A
2014/0207224	12/2013	Simon	N/A	N/A
2014/0257168	12/2013	Gill	N/A	N/A
2014/0324128	12/2013	Gross	N/A	N/A
2015/0011927	12/2014	Hua	N/A	N/A
2015/0119898	12/2014	Desalles et al.	N/A	N/A
2016/0144164	12/2015	Sedighiani	N/A	N/A
2016/0331970	12/2015	Lozano	N/A	N/A
2017/0007823	12/2016	Gross	N/A	N/A
2017/0120053	12/2016	Fostick et al.	N/A	N/A
2017/0182317	12/2016	Gross et al.	N/A	N/A
2018/0071523	12/2017	Gross et al.	N/A	N/A

FOREIGN PATENT DOCUMENTS

Patent No.	Application Date	Country	CPC
94/05369	12/1993	WO	N/A
01/52931	12/2000	WO	N/A
01/85027	12/2000	WO	N/A
2001/085094	12/2000	WO	N/A
2004/066903	12/2003	WO	N/A
2005/105053	12/2004	WO	N/A
2006/064502	12/2005	WO	N/A
2006/064503	12/2005	WO	N/A
2006/090397	12/2005	WO	N/A
2006123346	12/2005	WO	N/A
2008/007369	12/2007	WO	N/A
2008/084477	12/2007	WO	N/A
2017/006327	12/2016	WO	N/A
2017/072769	12/2016	WO	N/A

OTHER PUBLICATIONS

A Notice of Allowance dated Nov. 22, 2021, which issued during the prosecution of U.S. Appl. No. 17/386,937. cited by applicant

A Corrected Notice of Allowability dated Jan. 18, 2022, which issued in U.S. Appl. No. 17/386,937. cited by applicant

Karran September E et201 al., 1 “The Amyloid cascade hypothesis for AD,” Nature Reviews Drug Discovery, vol. 10; 698-712,. cited by applicant

De La Torre JC, “Vascular Basis of Alzheimer's Pathogenesis,” Ann NY Acad Sci. 977:196-215 (Nov. 2002). cited by applicant

Weller RO et al., “Perivascular Drainage of Amyloid-b Peptides from the Brain and Its Failure in

Cerebral Amyloid Angiopathy and Alzheimer's Disease,” Brain Pathology 18 (Apr. 2008) 253-266. cited by applicant

Brief PubMed search for metal ions in Alzheimers. cited by applicant

An Office Action dated Sep. 27, 2016, which issued during the prosecution of U.S. Appl. No. 14/926,705. cited by applicant

An International Search Report and a Written Opinion both dated Aug. 7, 2008, which issued during the prosecution of Applicant's PCT/IL2007/000865. cited by applicant

An Office Action dated Mar. 29, 2013, which issued during the prosecution of U.S. Appl. No. 12/373,306. cited by applicant

An Office Action dated Oct. 31, 2011, which issued during the prosecution of U.S. Appl. No. 12/373,306. cited by applicant

An Office Action dated Oct. 1, 2012, which issued during the prosecution of U.S. Appl. No. 12/373,306. cited by applicant

Notice of Allowance dated Jul. 24, 2013, which issued during the prosecution of U.S. Appl. No. 12/373,306. cited by applicant

An Office Action dated Apr. 11, 2013, which issued during the prosecution of U.S. Appl. No. 13/663,757. cited by applicant

Notice of Allowance dated Oct. 28, 2013, which issued during the prosecution of U.S. Appl. No. 13/663,757. cited by applicant

Elixmann IM et al., “In-vitro evaluation of a drainage catheter with integrated bioimpedance electrodes to determine ventricular size,” Biomed Tech 2013; 58 (Suppl. 1) Sep. 2013 (2 pages total). cited by applicant

An Office Action dated Aug. 31, 2015, which issued during the prosecution of U.S. Appl. No. 13/872,794. cited by applicant

An Applicant Initiated Interview Summary dated Dec. 14, 2015, which issued during the prosecution of U.S. Appl. No. 13/872,794. cited by applicant

An Office Action dated Feb. 3, 2016, which issued during the prosecution of U.S. Appl. No. 13/872,794. cited by applicant

Notice of Allowance dated Dec. 9, 2016, which issued during the prosecution of U.S. Appl. No. 14/794,739. cited by applicant

An Applicant Initiated Interview Summary dated Feb. 25, 2016, which issued during the prosecution of U.S. Appl. No. 13/872,794. cited by applicant

An Office Action dated Jun. 15, 2016, which issued during the prosecution of U.S. Appl. No. 13/872,794. cited by applicant

An International Search Report and a Written Opinion both dated Oct. 20, 2016, which issued during the prosecution of Applicant's PCT/IL2016/050728. cited by applicant

An International Search Report and a Written Opinion both dated Sep. 2, 2008, which issued during the prosecution of Applicant's PCT/IL2008/000038. cited by applicant

An International Search Report and a Written Opinion both dated Jan. 26, 2017, which issued during the prosecution of Applicant's PCT/IL2016/051161. cited by applicant

An International Preliminary Report on Patentability dated Dec. 22, 2009, which issued during the prosecution of Applicant's PCT/IL2008/000038. cited by applicant

An Office Action dated May 26, 2017, which issued during the prosecution of U.S. Appl. No. 15/453,290. cited by applicant

An International Preliminary Report on Patentability dated Apr. 7, 2009, which issued during the prosecution of Applicant's PCT/IL2007/000865. cited by applicant

Loutzenhiser, “Membrane Potential measurements in renal afferent and efferent arterioles: actions of Angiotensin II”, AJP—Renal Physiol Aug. 1, 1997 vol. 273 No. 2 F307-F314. cited by applicant

U.S. Appl. No. 60/830,717, filed Jul. 12, 2006. cited by applicant

Dao-Sheng Liu et al., “Activation of Na⁺ and K⁺ Pumping Modes of (Na,K)-ATPase by an

Oscillating Electric Field,” The Journal of Biological Chemistry, vol. 265. No. 13, May 5, 1990. (pp. 7260-7267). cited by applicant

Robert F. Service.. “Electric fields deliver drugs into tumors.” <http://news.sciencemag.org>. Feb. 4, 2015. (5 Pages Total). cited by applicant

Vernengo J, “Injectable Bioadhesive Hydrogels for Nucleus Pulposus Replacement and Repair of the Damaged Intervertebral Disc: A Thesis,” Drexel University (Jan. 2007). cited by applicant

Urban JPG et al., “The nucleus of the intervertebral disc from development to degeneration,” American Zoologist 40(1): 53-61 (2000). cited by applicant

Cheung KMC et al., “Intervertebral disc regeneration by use of autologous mesenchymal stem cells, an experimental model in rabbits,” Abstract from the SRS 2004 Annual Meeting. cited by applicant

An Office Action dated Jan. 16, 2015, which issued during the prosecution of U.S. Appl. No. 13/872,794. cited by applicant

“Implanting ultrathin electrodes in soft tissue,” <https://www.youtube.com/watch?v=FaBYQ68JIM8&spfreload=10>, Published on Oct. 10, 2014 (excerpts from video). cited by applicant

Iatridis JC et al., “Influence of fixed charge density magnitude and distribution on the intervertebral disc: applications of a poroelastic and chemical electric (PEACE) model,” J Biomech Eng. 125(1):12-24 (Feb. 2003). cited by applicant

Freemont TJ et al., “Degeneration of intervertebral discs: current understanding of cellular and molecular events, and implications for novel therapies,” Expert Reviews in Molecular Biology, Mar. 29, 2001 (Cambridge University Press). cited by applicant

An Office Action dated Sep. 12, 2011, which issued during the prosecution of U.S. Appl. No. 12/373,306. cited by applicant

An Office Action dated Jul. 24, 2017, which issued during the prosecution of U.S. Appl. No. 14/982,187. cited by applicant

An International Search Report and a Written Opinion both dated Mar. 10, 2017, which issued during the prosecution of Applicant's PCT/IL2016/051363. cited by applicant

An Office Action dated Apr. 25, 2018, which issued during the prosecution of U.S. Appl. No. 15/637,330. cited by applicant

“ActivL® Artificial Disc” Patient Information, Aesculap ImplantSystems, LLC, Jun. 2015, pp. 1-49. cited by applicant

“Hydropneumatic suspension,” Wikipedia, downloaded May 19, 2019 from url https://en.wikipedia.org/wiki/Hydropneumatic_suspension, pp. 1-11. cited by applicant

“Charite Artificial Disc Patient Information”, DePuy Spine, Inc., Oct. 2004, pp. 34-41. cited by applicant

“ProDisc C Disc Replacement. For single level spinal arthroplasty from C3 to C7. Technique Guide”, Centinel Spine, Sep. 2017, pp. 1-32. cited by applicant

“M6-C Artificial Cervical Disc, The Natural Choice for Cervical Disc Replacement”, Orthofix Holdings, Inc., Mar. 2019, pp. 1-6. cited by applicant

Gornet, Matthew F. et al., “MAVERICK Total Disc Replacement,” Plastric Surgery Key, <https://plasticsurgerykey.com/maverick-total-disc-replacement/>, downloaded from url on Dec. 1, 2019, pp. 1-7. cited by applicant

“Kineflex Lumbar Disc”, Southern Medical, <http://www.southmed.co.za/kineflexlumbar-disc-72>, downloaded from url on Dec. 2, 2019, pp. 1-2. cited by applicant

Lauweryns, Philippe, “Design and Surgical Technique of the FlexiCore Lumbar Artificial Disc,” Neupsy Key, <https://neupsykey.com/design-and-surgicaltechnique-of-the-flexicore-lumbar-artificial-disc/>, downloaded from url on Dec. 1, 2019, pp. 1-3. cited by applicant

Szpalski M et al., “Spine arthroplasty: a historical review,” Eur Spine J, vol. 11, Suppl. 2, 2002, pp. S65-S84. cited by applicant

Notice of Allowance dated Dec. 13, 2017, which issued during the prosecution of U.S. Appl. No. 15/263,910. cited by applicant
Corrected Notice of Allowability dated Jan. 3, 2018, which issued during the prosecution of U.S. Appl. No. 15/263,910. cited by applicant
An Invitation to pay additional fees dated Dec. 12, 2017, which issued during the prosecution of Applicant's PCT/IL2017/051032. cited by applicant
An International Search Report and a Written Opinion both dated Feb. 12, 2018, which issued during the prosecution of Applicant's PCT/IL2017/051032. cited by applicant
Notice of Allowance dated May 7, 2021, which issued during the prosecution of U.S. Appl. No. 16/332,606. cited by applicant

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS (1) The present application is a divisional of U.S. application Ser. No. 17/386,937, filed Jul. 28, 2021, which is a continuation of U.S. application Ser. No. 16/332,606, filed Mar. 12, 2019. now U.S. Pat. No. 11,097,098, which is the U.S. national stage of International Application PCT/IL2017/051032, filed Sep. 13, 2017, which is a continuation of U.S. application Ser. No. 15/263,910, filed Sep. 13, 2016, now U.S. Pat. No. 9,950,156, which is assigned to the assignee of the present application and is incorporated herein by reference.

FIELD OF THE APPLICATION

(1) The present invention relates generally to therapeutic electrical techniques.

BACKGROUND OF THE APPLICATION

(2) The intervertebral discs form cartilaginous joints between the end plates of vertebrae to provide shock absorption. The discs include two main regions: the nucleus pulposus, which is an inner, soft and highly hydrated structure, and the annulus fibrosus, which is a strong structure including lamellae (concentric sheets of collagen fibers), which surrounds the nucleus. The three major constituents of the discs are water, fibrillar collagens, and aggrecan. The proportion of these components varies across the disc, with the nucleus having a higher concentration of aggrecan and water and a lower collagen content than other regions of the disc. The loss of water content, particularly in the nucleus pulposus, is associated with disc degeneration, and with a decrease in disc height and abnormal loading of other spinal structures.

(3) U.S. Pat. No. 8,577,469 to Gross, which is assigned to the assignee of the present application and is incorporated herein by reference, describes apparatus for treating an intervertebral disc of a subject. The apparatus includes a first electrode, configured to be inserted into a nucleus pulposus of the disc, and a second electrode, configured to be placed outside of the nucleus pulposus, in a vicinity of the nucleus pulposus. A control unit is configured to drive a current between the first and second electrodes, and to configure the current to electroosmotically drive fluid between inside and outside the nucleus pulposus. Other embodiments are also described

(4) US Patent Application Publication 2005/0277996 to Podhajsky describes a method for reducing intervertebral pressure, including providing an electrode, having proximal and distal ends, and a generator, which is operatively connected to the proximal end of the electrode, and is configured to supply radiofrequency current thereto. The method also includes inserting at least a portion of the distal end of the electrode into the nucleus pulposus of an intervertebral disc and activating the

generator to heat the nucleus pulposus. The electrode may be inserted into the intervertebral disc through its first lateral side and/or its second lateral side, and may be substantially parallel to the major or minor axis of the nucleus pulposus.

SUMMARY OF THE APPLICATION

(5) In some embodiments of the present invention, an intervertebral-disc-treatment system is provided for treating an intervertebral disc of a subject. The intervertebral-disc-treatment system comprises (a) an electrode, which is configured to be implanted partially within a nucleus pulposus of the disc, and (b) one or more extra-pulposus exposed electrode surfaces, which are configured to be implanted outside the nucleus pulposus, in electrical communication with the disc, in a vicinity of an external surface of an annulus fibrosus of the disc. The intervertebral-disc-treatment system further comprises implantable or external control circuitry, which is typically electrically coupled, by one or more electrode leads, to the exposed electrode surfaces and the electrode.

(6) For some applications, the electrode comprises a wire, which typically has a wire diameter of between 50 and 150 microns, such as 100 microns. The wire includes: a non-electrically-insulated current-application longitudinal segment, which, in the absence of any applied forces, is helically coiled and typically has (a) a mean pitch of between 1.1 and 2 times the wire diameter, and (b) an entire longitudinal length of between 5 and 35 mm, and an electrically-insulated lead longitudinal segment, which typically has an entire longitudinal length of between 5 and 80 mm, in the absence of any applied forces.

(7) For some applications, the current-application longitudinal segment has an outer coil diameter of between 3 and 7 times the wire diameter, in the absence of any applied forces.

(8) Typically, because of the above-mentioned dimensions, including the pitch, the current-application longitudinal segment is highly flexible (both in its ability to bend and its ability to longitudinally stretch and contract), which reduces the application of forces on the intervertebral disc, and thus generally causes no or less trauma to the intervertebral disc, including the nucleus pulposus, than a more rigid electrode might cause, particularly over time during repeated motion of the disc.

(9) For some applications, the lead longitudinal segment is coiled, in the absence of any applied forces. For some applications, the lead longitudinal segment has one or more of the following dimensions, in the absence of any applied forces: (a) a mean pitch that is greater than the mean pitch $P_{sub.1}$ of the current-application longitudinal segment, (b) a mean pitch of between 2 and 3 times the wire diameter, (c) an outer coil diameter that equals between 90% and 110% of the outer coil diameter of the current-application longitudinal segment, and/or (d) an outer coil diameter of between 3 and 7 times the wire diameter. Typically, providing the above-mentioned pitches enables the thorough application of insulation to the electrically-insulated lead longitudinal segment, particularly if a vapor deposition process is used to apply the insulation (e.g., parylene). In addition, the lead longitudinal segment may have a greater pitch than the current-application longitudinal segment in part because the lead longitudinal segment generally does not require as much flexibility as the current-application longitudinal segment.

(10) For some applications, the wire further includes an intermediate longitudinal segment, which (a) is longitudinally between the current-application longitudinal segment and the lead longitudinal segment, (b) typically has an entire longitudinal length of between 1 and 6 mm, in the absence of any applied forces, and (c) in the absence of any applied forces, either (i) is coiled, and has a mean pitch greater than the outer coil diameter of the current-application longitudinal segment, or (ii) is not coiled. For some applications, the intermediate longitudinal segment is electrically insulated along at least a longitudinal portion of the intermediate longitudinal segment. For some applications in which the intermediate longitudinal segment is coiled, the intermediate longitudinal segment has one or more of the following dimensions, in the absence of any applied forces: (a) a mean pitch of between 125% and 250% of the outer coil diameter of the current-application longitudinal segment, (b) a mean pitch of between 5 and 20 times the wire diameter, (c)

a mean pitch P.sub.3 that equals (i) at least the mean pitch of the current-application longitudinal segment, and (ii) no more than the mean pitch of the lead longitudinal segment, (d) an outer coil diameter equal to between 90% and 110% of the outer coil diameter of the current-application longitudinal segment, and/or (e) an outer coil diameter of between 3 and 7 times the wire diameter. (11) For some applications, the wire further includes a pin-connector longitudinal segment. The lead longitudinal segment is longitudinally between the current-application longitudinal segment and the pin-connector longitudinal segment. For these applications, the intervertebral-disc-treatment system typically comprises an electrode assembly, which comprises the electrode and a pin, which is mechanically fixed to and in electrical communication with the pin-connector longitudinal segment. Typically, the area surrounding the pin and the pin-connector longitudinal segment is electrically insulated, such as by silicone.

(12) At least a portion of the electrode is implanted in a body of a subject, typically such that: the current-application longitudinal segment is disposed at least partially in the intervertebral disc of the subject, typically entirely in the intervertebral disc (either entirely in the nucleus pulposus of the intervertebral disc, or partially in a nucleus pulposus of the intervertebral disc and partially in the annulus fibrosus of the intervertebral disc), at least a portion of the lead longitudinal segment is disposed in the body of the subject outside the intervertebral disc, and if provided, the intermediate longitudinal segment is disposed at least in part in the annulus fibrosus of the intervertebral disc; providing the relatively large pitch for the intermediate longitudinal segment, or not coiling the intermediate longitudinal segment, may reduce the likelihood that the portion of the intermediate longitudinal segment that is within the annulus fibrosus forms a tunnel, which might cause leakage of fluid from the nucleus pulposus.

(13) In some applications of the present invention, a method of implanting at least a portion of an electrode in a body of a subject is provided, the method comprising: inserting a hollow insertion needle into an intervertebral disc of the subject; aligning an electrode loader with the hollow insertion needle, while the electrode is preloaded in a hollow loader needle of the electrode loader, and a loader stylet of the electrode loader is preloaded partially in the hollow loader needle and disposed such a distal end of the loader stylet abuts a proximal end of the electrode; connecting the electrode loader to the hollow insertion needle, such that a distal end of the hollow loader needle abuts a proximal end of the hollow insertion needle; advancing the electrode into the hollow insertion needle by advancing the loader stylet distally within the hollow loader needle so that the loader stylet pushes the electrode distally from the hollow loader needle into the insertion needle; aligning a needle-withdrawal handle and a handle stylet thereof with the hollow insertion needle; connecting the needle-withdrawal handle to the hollow insertion needle; proximally withdrawing the hollow insertion needle from the intervertebral disc, while holding the needle-withdrawal handle stationary, and while the handle stylet abuts the proximal end of the electrode, thereby preventing proximal motion of the electrode; and proximally withdrawing the needle-withdrawal handle, thereby releasing the electrode and leaving the electrode implanted partially in the intervertebral disc and partially in the body of the subject outside the intervertebral disc.

(14) For some applications, inserting the hollow insertion needle comprises limiting a depth of penetration of the hollow insertion needle into the intervertebral disc by inserting the hollow insertion needle into the intervertebral disc until a spacer, through which the hollow insertion needle passes, contacts an external surface of an annulus fibrosis of the intervertebral disc. For some applications, connecting the needle-withdrawal handle to the hollow insertion needle comprises connecting the needle-withdrawal handle to the spacer. For some applications, proximally withdrawing the needle-withdrawal handle comprises proximally withdrawing the needle-withdrawal handle while the spacer is attached to the needle-withdrawal handle.

(15) For some applications, the hollow insertion needle is axially fixed to a needle-connection fitting. Connecting the needle-withdrawal handle to the hollow insertion needle comprises disposing (a) the needle-connection fitting within the needle-withdrawal handle, and (b) a distal

end of the handle stylet within the needle-connection fitting abutting the proximal end of the electrode. Proximally withdrawing the hollow insertion needle comprises proximally withdrawing the needle-connection fitting within the needle-withdrawal handle, while the needle-withdrawal handle is held stationary.

(16) In some applications of the present invention, the intervertebral-disc-treatment system comprises: at least one intra-pulposus exposed electrode surface, which is configured to be implanted in the nucleus pulposus of the intervertebral disc; a plurality of extra-pulposus exposed electrode surfaces, which are configured to be implanted outside the nucleus pulposus, in electrical communication with the intervertebral disc; and the control circuitry, which is (a) electrically coupled to the at least one intra-pulposus exposed electrode surface and the plurality of the extra-pulposus exposed electrode surfaces, (b) configured to separately control at least two of the plurality of the extra-pulposus exposed electrode surfaces; for example, the control circuitry may be electrically coupled to the extra-pulposus exposed electrode surfaces separately via separate electrical conductors.

(17) Providing the plurality of separately-controllable the extra-pulposus exposed electrode surfaces distributes the generation of hydrogen, thereby reducing any local build-up of hydrogen at any single electrode surface.

(18) For some applications, the control circuitry is configured to: repeatedly assume a pressure-increasing mode of operation in alternation with an oxygen-generating mode of operation, in the pressure-increasing mode of operation, (a) configure the at least one intra-pulposus exposed electrode surface to be a cathode, and one or more of the plurality of the extra-pulposus exposed electrode surfaces to be one or more respective anodes, and (b) electroosmotically drive fluid into the nucleus pulposus to increase pressure in the intervertebral disc, by applying a first mean voltage of less than 1.23 V (sometimes known in the art as the “electrolysis voltage”) between the at least one intra-pulposus exposed electrode surface and the one or more of the plurality of the extra-pulposus exposed electrode surfaces, and in the oxygen-generating mode of operation, (a) configure the at least one intra-pulposus exposed electrode surface to be an anode, and the plurality of the extra-pulposus exposed electrode surfaces to be a respective plurality of cathodes, and (b) generate oxygen within the nucleus pulposus by electrolysis, by applying a second mean voltage of at least 1.23 V between the at least one intra-pulposus exposed electrode surface and the plurality of the extra-pulposus exposed electrode surfaces.

(19) The increase in fluid in the nucleus pulposus during the pressure-increasing mode of operation generally treats or prevents further degeneration of the disc caused at least in part by loss of fluid. The applied current may also help introduce nutritional substances into the disc. The generation of oxygen within the nucleus pulposus during the oxygen-generating mode generally treats hypoxia, which, if untreated, sometimes causes disc degeneration. The generation of oxygen may also improve glucose metabolism, while reducing lactic acid generation.

(20) For some applications, the control circuitry is configured to apply direct current, e.g., with an average amplitude of between 1 and 5 mA. For some applications, the control unit is configured to apply the direct current as a series of pulses. For some applications, the control unit is configured to apply the direct current as the series of pulses with a duty cycle of between 1% and 50%.

(21) Typically, the control circuitry is configured to separately control all of the plurality of the extra-pulposus exposed electrode surfaces; for example, the control circuitry may be electrically coupled to the extra-pulposus exposed electrode surfaces separately via separate electrical conductors.

(22) For some applications, the control circuitry is configured, during a period of time, to assume (a) the pressure-increasing mode of operation at least 10 times for an aggregate first duration and (b) the oxygen-generating mode of operation at least 10 times for an aggregate second duration that is less than 10% of the aggregate first duration. By way of example and not limitation, the control circuitry may assume the oxygen-generating mode of operation for a few seconds every hour, and

assume the pressure-increasing mode of operation at other times. Although the control circuitry, when in the oxygen-generating mode of operation, may electroosmotically drive fluid out of the nucleus pulposus and thus decrease pressure in the intervertebral disc, because the aggregate second duration is so much less than the aggregate first duration, the aggregate effect of the application of voltages is an increase in pressure in the intervertebral disc.

(23) For some applications, the control circuitry is configured to, in the oxygen-generating mode of operation, generate oxygen within the nucleus pulposus by electrolysis, by applying the second mean voltage between the at least one intra-pulposus exposed electrode surface and respective different subsets of the plurality of the extra-pulposus exposed electrode surfaces at respective different times. For some applications, each of the subsets consists of exactly one of the plurality of the extra-pulposus exposed electrode surfaces. Activating the extra-pulposus exposed electrode surfaces at different times further distributes the generation of hydrogen, thereby further reducing any local build-up of hydrogen at any single electrode surface.

(24) In some applications of the present invention, the intervertebral-disc-treatment system comprises: at least one intra-pulposus exposed electrode surface, which is configured to be implanted in the nucleus pulposus of the intervertebral disc; one or more the extra-pulposus exposed electrode surfaces, which (a) are configured to be implanted outside the nucleus pulposus, in electrical communication with the intervertebral disc, and (b) have an aggregate electrically-exposed surface area of at least 3 cm², such as at least 4 cm², e.g., at least 5 cm²; and the control circuitry, which is electrically coupled to the at least one intra-pulposus exposed electrode surface and one or more the extra-pulposus exposed electrode surfaces.

(25) For some applications, the control circuitry is configured to: repeatedly assume a pressure-increasing mode of operation in alternation with an oxygen-generating mode of operation, in the pressure-increasing mode of operation, (a) configure the at least one intra-pulposus exposed electrode surface to be a cathode, and the one or more extra-pulposus exposed electrode surfaces to be one or more respective anodes, and (b) electroosmotically drive fluid into the nucleus pulposus to increase pressure in the intervertebral disc, by applying a first mean voltage of less than 1.23 V between the at least one intra-pulposus exposed electrode surface and the one or more extra-pulposus exposed electrode surfaces, and in the oxygen-generating mode of operation, (a) configure the at least one intra-pulposus exposed electrode surface to be an anode, and the one or more extra-pulposus exposed electrode surfaces to be a respective plurality of cathodes, and (b) generate oxygen within the nucleus pulposus by electrolysis, by applying a second mean voltage of at least 1.23 V between the at least one intra-pulposus exposed electrode surface and the one or more extra-pulposus exposed electrode surfaces.

(26) The increase in fluid in the nucleus pulposus during the pressure-increasing mode of operation generally treats or prevents further degeneration of the disc caused at least in part by loss of fluid. The applied current may also help introduce nutritional substances into the disc. The generation of oxygen within the nucleus pulposus during the oxygen-generating mode generally treats hypoxia, which, if untreated, sometimes causes disc degeneration. The generation of oxygen may also improve glucose metabolism, while reducing lactic acid generation. Providing the relatively large aggregate electrically-exposed surface area of at least 3 cm² distributes the generation of hydrogen, thereby reducing any local build-up of hydrogen at the electrode-tissue interface.

(27) For some applications, the intervertebral-disc-treatment system comprises exactly one extra-pulposus exposed electrode surface having an electrically-exposed surface area of at least 3 cm². Alternatively, an external surface of a can of the control circuitry serves as the extra-pulposus exposed electrode surface. For other applications, the intervertebral-disc-treatment system comprises a plurality of the extra-pulposus exposed electrode surfaces.

(28) For some applications, the control circuitry is configured to apply direct current, e.g., with an average amplitude of between 1 and 5 mA. For some applications, the control unit is configured to apply the direct current as a series of pulses. For some applications, the control unit is configured to

apply the direct current as the series of pulses with a duty cycle of between 1% and 50%.

(29) For some applications, the control circuitry is configured, during a period of time, to assume (a) the pressure-increasing mode of operation at least 10 times for an aggregate first duration and (b) the oxygen-generating mode of operation at least 10 times for an aggregate second duration that is less than 10% of the aggregate first duration. By way of example and not limitation, the control circuitry may assume the oxygen-generating mode of operation for a few seconds every hour, and assume the pressure-increasing mode of operation at other times. Although the control circuitry, when in the oxygen-generating mode of operation, may electroosmotically drive fluid out of the nucleus pulposus and thus decrease pressure in the intervertebral disc, because the aggregate second duration is so much less than the aggregate first duration, the aggregate effect of the application of voltages is an increase in pressure in the intervertebral disc.

(30) There is therefore provided, in accordance with an inventive concept 1 of the present invention, apparatus for treating an intervertebral disc of a subject, the apparatus including:

(31) at least one intra-pulposus exposed electrode surface, which is configured to be implanted in a nucleus pulposus of the intervertebral disc;

(32) a plurality of extra-pulposus exposed electrode surfaces, which are configured to be implanted outside the nucleus pulposus, in electrical communication with the intervertebral disc; and

(33) control circuitry, which is (a) electrically coupled to the at least one intra-pulposus exposed electrode surface and the plurality of extra-pulposus exposed electrode surfaces, (b) configured to separately control at least two of the plurality of extra-pulposus exposed electrode surfaces, and (c) configured to: repeatedly assume a pressure-increasing mode of operation in alternation with an oxygen-generating mode of operation, in the pressure-increasing mode of operation, (a) configure the at least one intra-pulposus exposed electrode surface to be a cathode, and one or more of the plurality of extra-pulposus exposed electrode surfaces to be one or more respective anodes, and (b) electroosmotically drive fluid into the nucleus pulposus to increase pressure in the intervertebral disc, by applying a first mean voltage of less than 1.23 V between the at least one intra-pulposus exposed electrode surface and the one or more of the plurality of extra-pulposus exposed electrode surfaces, and in the oxygen-generating mode of operation, (a) configure the at least one intra-pulposus exposed electrode surface to be an anode, and the plurality of extra-pulposus exposed electrode surfaces to be a respective plurality of cathodes, and (b) generate oxygen within the nucleus pulposus by electrolysis, by applying a second mean voltage of at least 1.23 V between the at least one intra-pulposus exposed electrode surface and the plurality of extra-pulposus exposed electrode surfaces.

Inventive concept 2. The apparatus according to inventive concept 1, wherein the control circuitry is configured to separately control all of the plurality of extra-pulposus exposed electrode surfaces. Inventive concept 3. The apparatus according to inventive concept 1, wherein the first mean voltage is less than 500 mV. Inventive concept 4. The apparatus according to inventive concept 3, wherein the first mean voltage is less than 300 mV. Inventive concept 5. The apparatus according to inventive concept 1, wherein the second mean voltage is at least 2 V.

Inventive concept 6. The apparatus according to any one of inventive concepts 1-5, wherein the control circuitry is configured, during a period of time, to assume (a) the pressure-increasing mode of operation at least 10 times for an aggregate first duration and (b) the oxygen-generating mode of operation at least 10 times for an aggregate second duration that is less than 10% of the aggregate first duration. Inventive concept 7. The apparatus according to inventive concept 6, wherein the aggregate second duration is less than 5% of the aggregate first duration. Inventive concept 8. The apparatus according to inventive concept 7, wherein the aggregate second duration is less than 1% of the aggregate first duration. Inventive concept 9. The apparatus according to any one of inventive concepts 1-5, wherein the plurality of extra-pulposus exposed electrode surfaces includes at least three extra-pulposus exposed electrode surfaces. Inventive concept 10. The apparatus according to inventive concept 9, wherein the plurality of extra-pulposus exposed electrode surfaces includes up to 10 extra-pulposus exposed electrode surfaces. Inventive concept 11. The

apparatus according to any one of inventive concepts 1-5, wherein the control circuitry is configured to, in the oxygen-generating mode of operation, generate oxygen within the nucleus pulposus by electrolysis, by applying the second mean voltage between the at least one intra-pulposus exposed electrode surface and respective different subsets of the plurality of extra-pulposus exposed electrode surfaces at respective different times. Inventive concept 12. The apparatus according to inventive concept 11, wherein each of the subsets consists of exactly one of the plurality of extra-pulposus exposed electrode surfaces. Inventive concept 13. The apparatus according to any one of inventive concepts 1-5,

(34) wherein the apparatus further includes a sensor, which is configured to sense a parameter indicative of a quantity of the oxygen generated by the electrolysis, and

(35) wherein the control circuitry is configured to modulate the oxygen-generating mode of operation responsively to the sensed parameter. Inventive concept 14. The apparatus according to inventive concept 13, wherein the control circuitry is configured to modulate the oxygen-generating mode of operation by modulating a duration of one or more occurrences of the oxygen-generating mode of operation. Inventive concept 15. The apparatus according to inventive concept 13, wherein the control circuitry is configured to modulate the oxygen-generating mode of operation by modulating an electrical parameter of the second mean voltage. Inventive concept 16. The apparatus according to inventive concept 13, wherein the sensed parameter is an oxygen concentration in the nucleus pulposus. Inventive concept 17. The apparatus according to inventive concept 13, wherein the sensed parameter is a pH in the nucleus pulposus.

(36) There is further provided, in accordance with an inventive concept 18 of the present invention, apparatus for treating an intervertebral disc of a subject, the apparatus including: at least one intra-pulposus exposed electrode surface, which is configured to be implanted in a nucleus pulposus of the intervertebral disc; one or more extra-pulposus exposed electrode surfaces, which (a) are configured to be implanted outside the nucleus pulposus, in electrical communication with the intervertebral disc, and (b) have an aggregate electrically-exposed surface area of at least 1 cm²; and control circuitry, which is (a) electrically coupled to the at least one intra-pulposus exposed electrode surface and one or more extra-pulposus exposed electrode surfaces, and (b) configured to: repeatedly assume a pressure-increasing mode of operation in alternation with an oxygen-generating mode of operation, in the pressure-increasing mode of operation, (a) configure the at least one intra-pulposus exposed electrode surface to be a cathode, and the one or more extra-pulposus exposed electrode surfaces to be one or more respective anodes, and (b) electroosmotically drive fluid into the nucleus pulposus to increase pressure in the intervertebral disc, by applying a first mean voltage of less than 1.23 V between the at least one intra-pulposus exposed electrode surface and the one or more extra-pulposus exposed electrode surfaces, and in the oxygen-generating mode of operation, (a) configure the at least one intra-pulposus exposed electrode surface to be an anode, and the one or more extra-pulposus exposed electrode surfaces to be a respective plurality of cathodes, and (b) generate oxygen within the nucleus pulposus by electrolysis, by applying a second mean voltage of at least 1.23 V between the at least one intra-pulposus exposed electrode surface and the one or more extra-pulposus exposed electrode surfaces. Inventive concept 19. The apparatus according to inventive concept 18, wherein the aggregate electrically-exposed surface area is at least 3 cm². Inventive concept 20. The apparatus according to inventive concept 18, wherein the first mean voltage is less than 500 mV. Inventive concept 21. The apparatus according to inventive concept 20, wherein the first mean voltage is less than 300 mV. Inventive concept 22. The apparatus according to inventive concept 18, wherein the second mean voltage is at least 2 V. Inventive concept 23. The apparatus according to any one of inventive concepts 18-22, wherein the control circuitry is configured, during a period of time, to assume (a) the pressure-increasing mode of operation at least 10 times for an aggregate first duration and (b) the oxygen-generating mode of operation at least 10 times for an aggregate second duration that is less than 10% of the aggregate first duration. Inventive concept 24. The apparatus according to

inventive concept 23, wherein the aggregate second duration is less than 5% of the aggregate first duration. Inventive concept 25. The apparatus according to inventive concept 24, wherein the aggregate second duration is less than 1% of the aggregate first duration. Inventive concept 26. The apparatus according to any one of inventive concepts 18-22, wherein the apparatus includes exactly one extra-pulposus exposed electrode surface having an electrically-exposed surface area of at least 1 cm². Inventive concept 27. The apparatus according to inventive concept 26, wherein the surface area of the of the exactly one extra-pulposus exposed electrode surface is at least 3 cm². Inventive concept 28. The apparatus according to any one of inventive concepts 18-22, wherein the apparatus includes a plurality of extra-pulposus exposed electrode surfaces. Inventive concept 29. The apparatus according to inventive concept 18,

(37) wherein the apparatus further includes a sensor, which is configured to sense a parameter indicative of a quantity of the oxygen generated by the electrolysis, and

(38) wherein the control circuitry is configured to modulate the oxygen-generating mode of operation responsively to the sensed parameter. Inventive concept 30. The apparatus according to inventive concept 29, wherein the control circuitry is configured to modulate the oxygen-generating mode of operation by modulating a duration of one or more occurrences of the oxygen-generating mode of operation. Inventive concept 31. The apparatus according to inventive concept 29, wherein the control circuitry is configured to modulate the oxygen-generating mode of operation by modulating an electrical parameter of the second mean voltage. Inventive concept 32. The apparatus according to inventive concept 29, wherein the sensed parameter is an oxygen concentration in the nucleus pulposus. Inventive concept 33. The apparatus according to inventive concept 29, wherein the sensed parameter is a pH in the nucleus pulposus.

(39) There is still further provided, in accordance with an inventive concept 34 of the present invention, a method for treating an intervertebral disc of a subject, the method including: implanting at least one intra-pulposus exposed electrode surface in a nucleus pulposus of the intervertebral disc; implanting a plurality of extra-pulposus exposed electrode surfaces outside the nucleus pulposus, in electrical communication with the intervertebral disc; and activating control circuitry, which is (a) electrically coupled to the at least one intra-pulposus exposed electrode surface and the plurality of extra-pulposus exposed electrode surfaces, and (b) configured to separately control at least two of the plurality of extra-pulposus exposed electrode surfaces, to: repeatedly assume a pressure-increasing mode of operation in alternation with an oxygen-generating mode of operation, in the pressure-increasing mode of operation, (a) configure the at least one intra-pulposus exposed electrode surface to be a cathode, and one or more of the plurality of extra-pulposus exposed electrode surfaces to be one or more respective anodes, and (b) electroosmotically drive fluid into the nucleus pulposus to increase pressure in the intervertebral disc, by applying a first mean voltage of less than 1.23 V between the at least one intra-pulposus exposed electrode surface and the one or more of the plurality of extra-pulposus exposed electrode surfaces, and in the oxygen-generating mode of operation, (a) configure the at least one intra-pulposus exposed electrode surface to be an anode, and the plurality of extra-pulposus exposed electrode surfaces to be a respective plurality of cathodes, and (b) generate oxygen within the nucleus pulposus by electrolysis, by applying a second mean voltage of at least 1.23 V between the at least one intra-pulposus exposed electrode surface and the plurality of extra-pulposus exposed electrode surfaces.

(40) There is additionally provided, in accordance with an inventive concept 35 of the present invention, a method for treating an intervertebral disc of a subject, the method including: implanting at least one intra-pulposus exposed electrode surface in a nucleus pulposus of the intervertebral disc; implanting one or more extra-pulposus exposed electrode surfaces outside the nucleus pulposus, in electrical communication with the intervertebral disc, the one or more extra-pulposus exposed electrode surfaces having an aggregate electrically-exposed surface area of at least 1 cm²; and activating control circuitry, which is electrically coupled to the at least one intra-

pulposus exposed electrode surface and one or more extra-pulposus exposed electrode surfaces, to: repeatedly assume a pressure-increasing mode of operation in alternation with an oxygen-generating mode of operation, in the pressure-increasing mode of operation, (a) configure the at least one intra-pulposus exposed electrode surface to be a cathode, and the one or more extra-pulposus exposed electrode surfaces to be one or more respective anodes, and (b) electroosmotically drive fluid into the nucleus pulposus to increase pressure in the intervertebral disc, by applying a first mean voltage of less than 1.23 V between the at least one intra-pulposus exposed electrode surface and the one or more extra-pulposus exposed electrode surfaces, and in the oxygen-generating mode of operation, (a) configure the at least one intra-pulposus exposed electrode surface to be an anode, and the one or more extra-pulposus exposed electrode surfaces to be a respective plurality of cathodes, and (b) generate oxygen within the nucleus pulposus by electrolysis, by applying a second mean voltage of at least 1.23 V between the at least one intra-pulposus exposed electrode surface and the one or more extra-pulposus exposed electrode surfaces.

(41) There is yet additionally provided, in accordance with an inventive concept 36 of the present invention, apparatus including an electrode, which includes a wire that (a) has a wire diameter of between 75 and 125 microns and (b) includes: a non-electrically-insulated current-application longitudinal segment, which, in the absence of any applied forces, is coiled and has (i) a mean pitch of between 1.1 and 2 times the wire diameter, and (ii) an entire longitudinal length of between 5 and 35 mm, and an electrically-insulated lead longitudinal segment, which has an entire longitudinal length of between 5 and 80 mm, in the absence of any applied forces. Inventive concept 37. The apparatus according to inventive concept 36, wherein the mean pitch of the current-application longitudinal segment is between 1.2 and 1.8 times the wire diameter, in the absence of any applied forces. Inventive concept 38. The apparatus according to inventive concept 37, wherein the mean pitch of the current-application longitudinal segment is between 1.4 and 1.6 times the wire diameter, in the absence of any applied forces. Inventive concept 39. The apparatus according to inventive concept 36, wherein the wire diameter is between 90 and 110 microns. Inventive concept 40. The apparatus according to inventive concept 39, wherein the wire diameter of is between 95 and 105 microns. Inventive concept 41. The apparatus according to inventive concept 36, wherein the current-application longitudinal segment has an outer coil diameter of between 3 and 7 times the wire diameter, in the absence of any applied forces. Inventive concept 42. The apparatus according to inventive concept 41, wherein the outer coil diameter of the current-application longitudinal segment is between 4 and 6 times the wire diameter, in the absence of any applied forces. Inventive concept 43. The apparatus according to inventive concept 36, wherein the current-application longitudinal segment has an outer coil diameter of between 400 and 600 microns, in the absence of any applied forces. Inventive concept 44. The apparatus according to inventive concept 36, wherein the entire longitudinal length of the current-application longitudinal segment is between 10 and 25 mm, in the absence of any applied forces. Inventive concept 45. The apparatus according to inventive concept 36, wherein the entire longitudinal length of the electrically-insulated lead longitudinal segment is between 10 and 50 mm, in the absence of any applied forces. Inventive concept 46. The apparatus according to inventive concept 36, wherein, when a central longitudinal axis of a 3-mm long portion of the current-application longitudinal segment is curved in a plane only until all adjacent turns of the coiled wire of the portion touch one another, the central longitudinal axis of the portion has a radius of curvature of between 5 and 10 mm. Inventive concept 47. The apparatus according to inventive concept 36, wherein the current-application longitudinal segment extends to a distal end of the wire. Inventive concept 48. The apparatus according to inventive concept 36, wherein the electrically-insulated lead longitudinal segment is coated with parylene. Inventive concept 49. The apparatus according to inventive concept 36, wherein the lead longitudinal segment is not coiled, in the absence of any applied forces. Inventive concept 50. The apparatus according to any one of inventive concepts 36-49, wherein the lead longitudinal segment is coiled, in the absence of any applied forces. Inventive

concept 51. The apparatus according to inventive concept 50, wherein the lead longitudinal segment has a mean pitch that is greater than the mean pitch of the current-application longitudinal segment, in the absence of any applied forces. Inventive concept 52. The apparatus according to inventive concept 51, wherein the mean pitch of the lead longitudinal segment is at least 125% of the mean pitch of the current-application longitudinal segment, in the absence of any applied forces. Inventive concept 53. The apparatus according to inventive concept 52, wherein the mean pitch of the lead longitudinal segment is at least 150% of the mean pitch of the current-application longitudinal segment, in the absence of any applied forces. Inventive concept 54. The apparatus according to inventive concept 50, wherein the lead longitudinal segment has a mean pitch of between 2 and 3 times the wire diameter, in the absence of any applied forces. Inventive concept 55. The apparatus according to inventive concept 54, wherein the mean pitch of the lead longitudinal segment is between 2.2 and 2.8 times the wire diameter, in the absence of any applied forces. Inventive concept 56. The apparatus according to inventive concept 55, wherein the mean pitch of the lead longitudinal segment is between 2.4 and 2.6 times the wire diameter, in the absence of any applied forces. Inventive concept 57. The apparatus according to inventive concept 50, wherein the lead longitudinal segment has an outer coil diameter that equals between 90% and 110% of an outer coil diameter of the current-application longitudinal segment, in the absence of any applied forces. Inventive concept 58. The apparatus according to inventive concept 57, wherein the outer coil diameter of the lead longitudinal segment equals the outer coil diameter of the current-application longitudinal segment, in the absence of any applied forces. Inventive concept 59. The apparatus according to inventive concept 50, wherein the lead longitudinal segment has an outer coil diameter of between 3 and 7 times the wire diameter, in the absence of any applied forces. Inventive concept 60. The apparatus according to inventive concept 59, wherein the outer coil diameter of the lead longitudinal segment is between 4 and 6 times the wire diameter, in the absence of any applied forces. Inventive concept 61. The apparatus according to any one of inventive concepts 36-49, wherein the wire further includes an intermediate longitudinal segment, which (a) is longitudinally between the current-application longitudinal segment and the lead longitudinal segment, (b) has an entire longitudinal length of between 1 and 6 mm, in the absence of any applied forces, and (c) in the absence of any applied forces, either (i) is coiled, and has a mean pitch greater than an outer coil diameter of the current-application longitudinal segment or (ii) is not coiled. Inventive concept 62. The apparatus according to inventive concept 61, wherein the intermediate longitudinal segment is electrically insulated along at least a longitudinal portion of the intermediate longitudinal segment. Inventive concept 63. The apparatus according to inventive concept 61, wherein the intermediate longitudinal segment is not coiled, in the absence of any applied forces. Inventive concept 64. The apparatus according to inventive concept 61, wherein the entire longitudinal length of the intermediate longitudinal segment is between 2 and 4 mm, in the absence of any applied forces. Inventive concept 65. The apparatus according to inventive concept 61, wherein, in the absence of any applied forces, the intermediate longitudinal segment is coiled, and has the mean pitch greater than the outer coil diameter of the current-application longitudinal segment. Inventive concept 66. The apparatus according to inventive concept 65, wherein the mean pitch of the intermediate longitudinal segment is between 125% and 250% of the outer coil diameter of the current-application longitudinal segment, in the absence of any applied forces. Inventive concept 67. The apparatus according to inventive concept 65, wherein an outer coil diameter of the intermediate longitudinal segment equals between 90% and 110% of the outer coil diameter of the current-application longitudinal segment, in the absence of any applied forces. Inventive concept 68. The apparatus according to inventive concept 67, wherein the outer coil diameter of the intermediate longitudinal segment equals the outer coil diameter of the current-application longitudinal segment, in the absence of any applied forces. Inventive concept 69. The apparatus according to inventive concept 65, wherein the intermediate longitudinal segment has a mean pitch of between 5 and 20 times the wire diameter, in the absence of any applied forces.

Inventive concept 70. The apparatus according to inventive concept 69, wherein the mean pitch of the intermediate longitudinal segment is between 6 and 13 times the wire diameter, in the absence of any applied forces. Inventive concept 71. The apparatus according to inventive concept 70, wherein the mean pitch of the intermediate longitudinal segment is between 9 and 11 times the wire diameter, in the absence of any applied forces. Inventive concept 72. The apparatus according to inventive concept 65, wherein the intermediate longitudinal segment has an outer coil diameter of between 3 and 7 times the wire diameter, in the absence of any applied forces. Inventive concept 73. The apparatus according to inventive concept 72, wherein the outer coil diameter of the intermediate longitudinal segment is between 4 and 6 times the wire diameter, in the absence of any applied forces. Inventive concept 74. The apparatus according to any one of inventive concepts 36-49,

(42) wherein the wire further includes a partially non-electrically-insulated pin-connector longitudinal segment,

(43) wherein the lead longitudinal segment is longitudinally between the current-application longitudinal segment and the pin-connector longitudinal segment, and

(44) wherein the apparatus includes an electrode assembly, which includes the electrode and a pin, which is mechanically fixed to and in electrical communication with the pin-connector longitudinal segment. Inventive concept 75. The apparatus according to inventive concept 74, wherein the pin is shaped so as to define a lumen. Inventive concept 76. The apparatus according to inventive concept 75, wherein the pin-connector longitudinal segment is at least partially disposed in the lumen of the pin.

(45) Inventive concept 77. The apparatus according to inventive concept 76, wherein the pin is crimped on the pin-connector longitudinal segment.

(46) Inventive concept 78. The apparatus according to inventive concept 74, wherein the pin-connector longitudinal segment is not coiled, in the absence of any applied forces. Inventive concept 79. The apparatus according to inventive concept 78, wherein the pin-connector longitudinal segment is coaxial with a central longitudinal axis of the electrically-insulated lead longitudinal segment, in the absence of any applied forces. Inventive concept 80. The apparatus according to inventive concept 74, wherein the pin-connector longitudinal segment is coiled, in the absence of any applied forces. Inventive concept 81. The apparatus according to inventive concept 80, wherein the pin-connector longitudinal segment has a mean pitch that equals at least 125% of an outer coil diameter of the current-application longitudinal segment, in the absence of any applied forces. Inventive concept 82. The apparatus according to inventive concept 81, wherein the mean pitch of the pin-connector longitudinal segment is at between 125% and 250% of the outer coil diameter of the current-application longitudinal segment, in the absence of any applied forces. Inventive concept 83. The apparatus according to inventive concept 80, wherein the pin-connector longitudinal segment has an outer coil diameter that equals between 90% and 110% of an outer coil diameter of the current-application longitudinal segment, in the absence of any applied forces. Inventive concept 84. The apparatus according to inventive concept 83, wherein the outer coil diameter of the pin-connector longitudinal segment equals the outer coil diameter of the current-application longitudinal segment, in the absence of any applied forces. Inventive concept 85. The apparatus according to inventive concept 80, wherein the pin-connector longitudinal segment, in the absence of any applied forces, has (a) a mean pitch of between 5 and 20 times the wire diameter, and (b) an entire longitudinal length of between 1 and 4 mm. Inventive concept 86. The apparatus according to inventive concept 85, wherein the mean pitch of the pin-connector longitudinal segment is between 5 and 10 times the wire diameter, in the absence of any applied forces. Inventive concept 87. The apparatus according to inventive concept 80, wherein the pin-connector longitudinal segment has an outer coil diameter of between 3 and 7 times the wire diameter, in the absence of any applied forces. Inventive concept 88. The apparatus according to inventive concept 87, wherein the outer coil diameter of the pin-connector longitudinal segment is

between 4 and 6 times the wire diameter, in the absence of any applied forces. Inventive concept 89. The apparatus according to any one of inventive concepts 36-49, wherein the wire further includes an intermediate longitudinal segment, which (a) is longitudinally between the current-application longitudinal segment and the lead longitudinal segment, and (b) has an entire longitudinal length of between 1 and 6 mm, in the absence of any applied forces, and wherein the electrode further includes a water-permeable element that surrounds at least a longitudinal portion of the intermediate longitudinal segment. Inventive concept 90. The apparatus according to inventive concept 89, wherein the water-permeable element includes a membrane. Inventive concept 91. The apparatus according to inventive concept 89, wherein the water-permeable element includes an acid selected from the group consisting of: glycolic acid and lactic acid. Inventive concept 92. The apparatus according to inventive concept 89, wherein the intermediate longitudinal segment is coiled, in the absence of any applied forces. Inventive concept 93. The apparatus according to inventive concept 92, wherein the intermediate longitudinal segment, in the absence of any applied forces, has a mean pitch that equals (a) at least the mean pitch of the current-application longitudinal segment, and (b) no more than the mean pitch of the lead longitudinal segment. Inventive concept 94. The apparatus according to any one of inventive concepts 36-49, further including control circuitry, which is electrically coupled to the electrode, and which is configured to drive the electrode to apply a current.

(47) There is also provided, in accordance with an inventive concept 95 of the present invention, apparatus including an electrode, which includes a wire that (a) has a wire diameter of between 75 and 125 microns and (b) includes a non-electrically-insulated current-application longitudinal segment, which, in the absence of any applied forces, is coiled and has (i) a mean pitch of between 1.1 and 2 times the wire diameter, and (ii) an entire longitudinal length of between 5 and 35 mm.

(48) There is further provided, in accordance with an inventive concept 96 of the present invention, apparatus including an electrode, which includes a wire that includes: a non-electrically-insulated current-application longitudinal segment, which, in the absence of any applied forces, is coiled and has an entire longitudinal length of between 5 and 35 mm, an electrically-insulated lead longitudinal segment, which has an entire longitudinal length of between 5 and 80 mm, in the absence of any applied forces, and an intermediate longitudinal segment, which (a) is longitudinally between the current-application longitudinal segment and the lead longitudinal segment, (b) has an entire longitudinal length of between 1 and 6 mm, in the absence of any applied forces, and (c) in the absence of any applied forces, either (i) is coiled, and has a mean pitch greater than an outer coil diameter of the current-application longitudinal segment or (ii) is not coiled.

Inventive concept 97. The apparatus according to inventive concept 96, wherein the intermediate longitudinal segment is electrically insulated along at least a longitudinal portion of the intermediate longitudinal segment. Inventive concept 98. The apparatus according to inventive concept 96, wherein the intermediate longitudinal segment is not coiled, in the absence of any applied forces. Inventive concept 99. The apparatus according to inventive concept 96, wherein the wire has a wire diameter of between 75 and 125 microns. Inventive concept 100. The apparatus according to any one of inventive concepts 96-99, wherein, in the absence of any applied forces, the intermediate longitudinal segment is coiled, and has the mean pitch greater than the outer coil diameter of the current-application longitudinal segment. Inventive concept 101. The apparatus according to inventive concept 100, wherein the mean pitch of the intermediate longitudinal segment is between 125% and 250% of the outer coil diameter of the current-application longitudinal segment, in the absence of any applied forces. Inventive concept 102. The apparatus according to inventive concept 100, wherein an outer coil diameter of the intermediate longitudinal segment equals between 90% and 110% of the outer coil diameter of the current-application longitudinal segment, in the absence of any applied forces. Inventive concept 103. The apparatus according to inventive concept 100, wherein in the absence of any applied forces: the current-application longitudinal segment has a mean pitch of between 1.1 and 2 times a wire diameter of

the wire, and the intermediate longitudinal segment has a mean pitch of between 5 and 20 times the wire diameter. Inventive concept 104. The apparatus according to inventive concept 100, wherein the intermediate longitudinal segment has an outer coil diameter of between 3 and 7 times the wire diameter, in the absence of any applied forces.

(49) There is still further provided, in accordance with an inventive concept 105 of the present invention, a method including: providing an electrode, which includes a wire that (a) has a wire diameter of between 75 and 125 microns and (b) includes (i) a non-electrically-insulated current-application longitudinal segment, which, in the absence of any applied forces, is coiled and has (A) a mean pitch of between 1.1 and 2 times the wire diameter, and (B) an entire longitudinal length of between 5 and 35 mm, and (ii) an electrically-insulated lead longitudinal segment has an entire longitudinal length of between 5 and 80 mm, in the absence of any applied forces; and implanting at least a portion of the electrode in a body of a subject. Inventive concept 106. The method according to inventive concept 105, wherein implanting the at least a portion of the electrode includes implanting the at least a portion of the electrode such that: the current-application longitudinal segment is disposed at least partially in an intervertebral disc of the subject, and at least a portion of the lead longitudinal segment is disposed in the body of the subject outside the intervertebral disc. Inventive concept 107. The method according to inventive concept 106, wherein implanting the at least a portion of the electrode includes implanting the at least a portion of the electrode such that the current-application longitudinal segment is disposed entirely in the intervertebral disc. Inventive concept 108. The method according to inventive concept 107, wherein implanting the at least a portion of the electrode includes implanting the at least a portion of the electrode such that the current-application longitudinal segment is disposed entirely in a nucleus pulposus of the intervertebral disc. Inventive concept 109. The method according to inventive concept 107, wherein implanting the at least a portion of the electrode includes implanting the at least a portion of the electrode such that the current-application longitudinal segment is disposed partially in a nucleus pulposus of the intervertebral disc and partially in an annulus fibrosus of the intervertebral disc. Inventive concept 110. The method according to inventive concept 107, wherein the wire further includes an intermediate longitudinal segment, which (a) is longitudinally between the current-application longitudinal segment and the lead longitudinal segment, (b) has an entire longitudinal length of between 1 and 6 mm, in the absence of any applied forces, and (c) in the absence of any applied forces, either (i) is coiled, and has a mean pitch greater than an outer coil diameter of the current-application longitudinal segment or (ii) is not coiled, and wherein implanting the at least a portion of the electrode includes implanting the at least a portion of the electrode such that the intermediate longitudinal segment is disposed at least in part in an annulus fibrosus of the intervertebral disc.

(50) There is additionally provided, in accordance with an inventive concept 111 of the present invention, a method including: providing an electrode, which includes a wire that (a) has a wire diameter of between 75 and 125 microns and (b) includes a non-electrically-insulated current-application longitudinal segment, which, in the absence of any applied forces, is coiled and has (a) a mean pitch of between 1.1 and 2 times the wire diameter, and (b) an entire longitudinal length of between 5 and 35 mm; and implanting at least a portion of the electrode in a body of a subject.

(51) There is yet additionally provided, in accordance with an inventive concept 112 of the present invention, a method including: providing an electrode, which includes a wire that includes (a) a non-electrically-insulated current-application longitudinal segment, which, in the absence of any applied forces, is coiled and has an entire longitudinal length of between 5 and 35 mm, (b) an electrically-insulated lead longitudinal segment, which has an entire longitudinal length of between 5 and 80 mm, in the absence of any applied forces, and (c) an intermediate longitudinal segment, which (i) is longitudinally between the current-application longitudinal segment and the lead longitudinal segment, (ii) has an entire longitudinal length of between 1 and 6 mm, in the absence of any applied forces, and (iii) in the absence of any applied forces, either (A) is coiled, and has a

mean pitch greater than an outer coil diameter of the current-application longitudinal segment or (B) is not coiled; and implanting at least a portion of the electrode in a body of a subject.

(52) There is also provided, in accordance with an inventive concept 113 of the present invention, a method of manufacturing an electrode, the method including: providing an electrode, which includes a non-electrically-insulated wire that (A) has a wire diameter of between 75 and 125 microns and (B) includes: (a) a current-application longitudinal segment, which, in the absence of any applied forces, is coiled and has a mean pitch of between 1.1 and 2 times the wire diameter, (b) a lead longitudinal segment, and (c) an intermediate longitudinal segment, which (i) is longitudinally between the current-application longitudinal segment and the lead longitudinal segment, and (ii) in the absence of any applied forces, either (A) is coiled and has a mean pitch greater than an outer coil diameter of the current-application longitudinal segment or (B) is not coiled; providing a polymer boot that is shaped so as to define an enclosed space with an opening having a perimeter; masking a portion of the electrode by placing the boot on the portion of the electrode, such that (a) the current-application longitudinal segment is within the enclosed space, (b) the lead longitudinal segment is outside the boot, and (c) the intermediate longitudinal segment passes through the opening such that the perimeter of the opening forms a tight seal with at least a portion of the intermediate longitudinal segment; placing the boot and the electrode into a vapor deposition chamber; and applying a parylene coating to the lead longitudinal segment by vapor deposition within the vapor deposition chamber. Inventive concept 114. The method according to inventive concept 113, wherein providing the electrode includes providing the electrode in which the current-application longitudinal segment has an entire longitudinal length of between 5 and 35 mm, in the absence of any applied forces. Inventive concept 115. The method according to inventive concept 113, wherein providing the electrode includes providing the electrode in which the lead longitudinal segment has an entire longitudinal length of between 5 and 80 mm, in the absence of any applied forces. Inventive concept 116. The method according to inventive concept 113, wherein providing the electrode includes providing the electrode in which the intermediate longitudinal segment has an entire longitudinal length of between 1 and 6 mm, in the absence of any applied forces. Inventive concept 117. The method according to inventive concept 113, wherein providing the electrode includes providing the electrode in which, in the absence of any applied forces: the current-application longitudinal segment has an entire longitudinal length of between 5 and 35 mm, the lead longitudinal segment has an entire longitudinal length of between 5 and 80 mm, and the intermediate longitudinal segment has an entire longitudinal length of between 1 and 6 mm. Inventive concept 118. The method according to inventive concept 113, wherein providing the polymer boot includes providing a silicone boot.

(53) There is further provided, in accordance with an inventive concept 119 of the present invention, apparatus including an electrode and an implantation system for implanting at least a portion of the electrode in a body of a subject, the implantation system including: a hollow insertion needle configured to be inserted into an intervertebral disc of the subject; an electrode loader, which (a) includes (i) a hollow loader needle in which the electrode is configured to be loaded, and (ii) a loader stylet, which is configured to be loaded partially in the hollow loader needle and to be disposed such that a distal end of the loader stylet abuts a proximal end of the electrode, and (b) is configured to be connected to the hollow insertion needle, such that (i) a distal end of the hollow loader needle abuts a proximal end of the hollow insertion needle, and (ii) the loader stylet pushes the electrode distally from the hollow loader needle into the insertion needle when the loader stylet is distally advanced within the hollow loader needle; and a needle-withdrawal handle, which (a) includes a handle stylet, and (b) is configured (i) to be connected to the hollow insertion needle, (ii) while held stationary, to proximally withdraw the hollow insertion needle from the intervertebral disc while the handle stylet abuts the proximal end of the electrode, thereby preventing proximal motion of the electrode, and (iii) by being proximally withdrawn, to release the electrode and leave the electrode implanted partially in the intervertebral disc and

partially in the body of the subject outside the intervertebral disc. Inventive concept 120. The apparatus according to inventive concept 119, wherein the implantation system further includes a spacer, which is configured to contact an external surface of an annulus fibrosis of the intervertebral disc so as to limit a depth of penetration of the hollow insertion needle into the intervertebral disc, and wherein the hollow insertion needle passes through the spacer. Inventive concept 121. The apparatus according to inventive concept 120, wherein the needle-withdrawal handle is configured to be connected to the hollow insertion needle by connecting the needle-withdrawal handle to the spacer. Inventive concept 122. The apparatus according to inventive concept 121, wherein the needle-withdrawal handle is configured to be proximally withdrawn by proximally withdrawing the needle-withdrawal handle while the spacer is attached to the needle-withdrawal handle. Inventive concept 123. The apparatus according to inventive concept 119, wherein the implantation system further includes a needle-connection fitting, which is axially fixed to the hollow insertion needle, wherein the needle-withdrawal handle is configured such that when the needle-withdrawal handle is connected to the hollow insertion needle, the needle-connection fitting is disposed within the needle-withdrawal handle, and a distal end of the handle stylet is disposed within the needle-connection fitting abutting the proximal end of the electrode, and wherein the needle-connection fitting is configured to be proximally withdrawn within the needle-withdrawal handle, while the needle-withdrawal handle is held stationary, so as to proximally withdraw the hollow insertion needle.

(54) There is still further provided, in accordance with an inventive concept 124 of the present invention, a method of implanting at least a portion of an electrode in a body of a subject, the method including: inserting a hollow insertion needle into an intervertebral disc of the subject; aligning an electrode loader with the hollow insertion needle, while (a) the electrode is preloaded in a hollow loader needle of the electrode loader, and (b) a loader stylet of the electrode loader is preloaded partially in the hollow loader needle and disposed such that a distal end of the loader stylet abuts a proximal end of the electrode; connecting the electrode loader to the hollow insertion needle, such that a distal end of the hollow loader needle abuts a proximal end of the hollow insertion needle; advancing the electrode into the hollow insertion needle by advancing the loader stylet distally within the hollow loader needle so that the loader stylet pushes the electrode distally from the hollow loader needle into the insertion needle; aligning a needle-withdrawal handle and a handle stylet of the needle-withdrawal handle with the hollow insertion needle; connecting the needle-withdrawal handle to the hollow insertion needle; proximally withdrawing the hollow insertion needle from the intervertebral disc, while holding the needle-withdrawal handle stationary, and while the handle stylet abuts the proximal end of the electrode so as to prevent proximal motion of the electrode; and proximally withdrawing the needle-withdrawal handle, thereby releasing the electrode and leaving the electrode implanted partially in the intervertebral disc and partially in the body of the subject outside the intervertebral disc. Inventive concept 125. The method according to inventive concept 124, wherein inserting the hollow insertion needle includes limiting a depth of penetration of the hollow insertion needle into the intervertebral disc by inserting the hollow insertion needle into the intervertebral disc until a spacer, through which the hollow insertion needle passes, contacts an external surface of an annulus fibrosis of the intervertebral disc. Inventive concept 126. The method according to inventive concept 125, wherein connecting the needle-withdrawal handle to the hollow insertion needle includes connecting the needle-withdrawal handle to the spacer. Inventive concept 127. The method according to inventive concept 126, wherein proximally withdrawing the needle-withdrawal handle includes proximally withdrawing the needle-withdrawal handle while the spacer is attached to the needle-withdrawal handle. Inventive concept 128. The method according to inventive concept 124, wherein the hollow insertion needle is axially fixed to a needle-connection fitting, wherein connecting the needle-withdrawal handle to the hollow insertion needle includes disposing (a) the needle-connection fitting within the needle-withdrawal handle, and (b) a distal end of the handle

stylet within the needle-connection fitting abutting the proximal end of the electrode, and wherein proximally withdrawing the hollow insertion needle includes proximally withdrawing the needle-connection fitting within the needle-withdrawal handle, while the needle-withdrawal handle is held stationary.

(55) There is additionally provided, in accordance with an inventive concept 129 of the present invention, apparatus for treating an eye of a subject, the apparatus including: a first exposed electrode surface, which is configured to be implanted in a vitreous cavity of the eye; a second exposed electrode surface, which is configured to be implanted in a body of the subject at a site outside the vitreous cavity; and control circuitry, which is configured to: repeatedly assume a pressure-decreasing mode of operation in alternation with an oxygen-generating mode of operation, in both the pressure-decreasing mode of operation and the oxygen-generating mode of operation, configure the first exposed electrode surface to be an anode, and the second exposed electrode surface to be a cathode, in the pressure-decreasing mode of operation, electroosmotically drive fluid from the vitreous cavity to outside the vitreous cavity to decrease pressure in the vitreous cavity, by applying a first mean voltage of less than 1.23 V between the first and the second exposed electrode surfaces, and in the oxygen-generating mode of operation, generate oxygen within the vitreous cavity by electrolysis, by applying a second mean voltage of at least 1.23 V between the first and the second exposed electrode surfaces. Inventive concept 130. The apparatus according to inventive concept 129, wherein the site is within a Schlemm's canal of the subject, and the second exposed electrode surface is configured to be implanted within the Schlemm's canal, and wherein the control circuitry is configured to, in the pressure-decreasing mode of operation, electroosmotically drive the fluid from the vitreous cavity to the Schlemm's canal to decrease the pressure in the vitreous cavity, by applying the first mean voltage between the first and the second exposed electrode surfaces. Inventive concept 131. The apparatus according to inventive concept 129, wherein the first mean voltage is less than 500 mV. Inventive concept 132. The apparatus according to inventive concept 131, wherein the first mean voltage is less than 300 mV. Inventive concept 133. The apparatus according to inventive concept 129, wherein the second mean voltage is at least 2 V. Inventive concept 134. The apparatus according to any one of inventive concepts 129-133, wherein the control circuitry is configured, during a period of time, to assume (a) the pressure-decreasing mode of operation at least 10 times for an aggregate first duration and (b) the oxygen-generating mode of operation at least 10 times for an aggregate second duration that is less than 10% of the aggregate first duration. Inventive concept 135. The apparatus according to inventive concept 134, wherein the aggregate second duration is less than 5% of the aggregate first duration. Inventive concept 136. The apparatus according to inventive concept 135, wherein the aggregate second duration is less than 1% of the aggregate first duration. Inventive concept 137. The apparatus according to any one of inventive concepts 129-133, wherein the apparatus further includes a sensor, which is configured to sense a parameter indicative of a quantity of the oxygen generated by the electrolysis, and wherein the control circuitry is configured to modulate the oxygen-generating mode of operation responsively to the sensed parameter. Inventive concept 138. The apparatus according to inventive concept 137, wherein the control circuitry is configured to modulate the oxygen-generating mode of operation by modulating a duration of one or more occurrences of the oxygen-generating mode of operation. Inventive concept 139. The apparatus according to inventive concept 137, wherein the control circuitry is configured to modulate the oxygen-generating mode of operation by modulating an electrical parameter of the second mean voltage. Inventive concept 140. The apparatus according to inventive concept 137, wherein the sensed parameter is an oxygen concentration in the vitreous cavity. Inventive concept 141. The apparatus according to inventive concept 137, wherein the sensed parameter is a pH in the vitreous cavity. Inventive concept 142. The apparatus according to any one of inventive concepts 129-133, wherein the control circuitry is configured to: detect a pressure difference between the vitreous cavity and outside the vitreous cavity, and modulate the pressure-decreasing mode of operation

responsively to the pressure difference parameter in response to the detected pressure difference.

(56) There is yet additionally provided, in accordance with an inventive concept 143 of the present invention, a method for treating an eye of a subject, the method including: implanting a first exposed electrode surface in a vitreous cavity of the eye; implanting a second exposed electrode surface in a body of the subject at a site outside the vitreous cavity; and activating control circuitry to: repeatedly assume a pressure-decreasing mode of operation in alternation with an oxygen-generating mode of operation, in both the pressure-decreasing mode of operation and the oxygen-generating mode of operation, configure the first exposed electrode surface to be an anode, and the second exposed electrode surface to be a cathode, in the pressure-decreasing mode of operation, electroosmotically drive fluid from the vitreous cavity to outside the vitreous cavity to decrease pressure in the vitreous cavity, by applying a first mean voltage of less than 1.23 V between the first and the second exposed electrode surfaces, and in the oxygen-generating mode of operation, generate oxygen within the vitreous cavity by electrolysis, by applying a second mean voltage of at least 1.23 V between the first and the second exposed electrode surfaces. Inventive concept 144.

The method according to inventive concept 143, wherein the site is within a Schlemm's canal of the subject, and wherein implanting the second exposed electrode surface includes implanting the second exposed electrode surface within the Schlemm's canal, and wherein activating the control circuitry includes activating the control circuitry to, in the pressure-decreasing mode of operation, electroosmotically drive the fluid from the vitreous cavity to the Schlemm's canal to decrease the pressure in the vitreous cavity, by applying the first mean voltage between the first and the second exposed electrode surfaces.

(57) The present invention will be more fully understood from the following detailed description of embodiments thereof, taken together with the drawings, in which:

Description

BRIEF DESCRIPTION OF THE DRAWINGS

- (1) FIG. 1 is a schematic illustration of an intervertebral-disc-treatment system for treating an intervertebral disc of a subject, in accordance with an application of the present invention;
- (2) FIG. 2 is a schematic illustration of an electrode of the intervertebral-disc-treatment system of FIG. 1, in accordance with an application of the present invention;
- (3) FIG. 3A is a schematic illustration of another configuration of the electrode of FIG. 2, in accordance with an application of the present invention;
- (4) FIG. 3B is a schematic illustration of yet another configuration of the electrode of FIG. 2, in accordance with an application of the present invention;
- (5) FIG. 4 is a schematic illustration of a 3-mm long portion of a current-application longitudinal segment of the electrode of FIG. 2, FIG. 3A, or FIG. 3B, in accordance with an application of the present invention;
- (6) FIG. 5 is a schematic illustration of another configuration of the electrode of FIG. 2, in accordance with an application of the present invention;
- (7) FIG. 6 is a flow chart illustrating a method of manufacturing the electrode of FIG. 2, of FIG. 3A, or of FIG. 3B, in accordance with an application of the present invention;
- (8) FIG. 7 is a schematic illustration of a polymer boot used in the manufacturing method of FIG. 6, in accordance with an application of the present invention;
- (9) FIG. 8 is schematic illustrations of an implantation system for implanting at least a portion of the electrode of FIG. 2 in a body of a subject, in accordance with an application of the present invention;
- (10) FIGS. 9A-B are schematic illustrations of several components of the implantation system of FIG. 8, in accordance with an application of the present invention;

- (11) FIG. 10 is a flow chart illustrating a method of implanting at least a portion of the electrode of FIG. 2 in a body of a subject, in accordance with an application of the present invention;
- (12) FIGS. 11A-B are schematic illustrations of a needle insertion step of the implantation method of FIG. 10, in accordance with an application of the present invention;
- (13) FIG. 12 is a schematic illustration of an anti-coring stylet retraction step of the implantation method of FIG. 10, in accordance with an application of the present invention;
- (14) FIGS. 13A-C are schematic illustrations of an electrode loader alignment step of the implantation method of FIG. 10, in accordance with an application of the present invention;
- (15) FIGS. 14A-B are schematic illustrations of an electrode-loader connection step of the implantation method of FIG. 10, in accordance with an application of the present invention;
- (16) FIGS. 15A-B are schematic illustrations of an electrode advancement step of the implantation method of FIG. 10, in accordance with an application of the present invention;
- (17) FIGS. 16A-C are schematic illustrations of a needle-withdrawal-handle alignment step of the implantation method of FIG. 10, in accordance with an application of the present invention;
- (18) FIGS. 17A-B are schematic illustrations of a needle-withdrawal-handle connection step of the implantation method of FIG. 10, in accordance with an application of the present invention;
- (19) FIGS. 18A-C are schematic illustrations of an insertion-needle withdrawal step of the implantation method of FIG. 10, in accordance with an application of the present invention;
- (20) FIG. 19 is a schematic illustration of a needle-withdrawal-handle withdrawal step of the implantation method of FIG. 10, in accordance with an application of the present invention;
- (21) FIG. 20 is a schematic illustration of the electrode of FIG. 2 implanted partially in the disc and partially in the body of the subject outside the disc, in accordance with an application of the present invention;
- (22) FIG. 21 is a schematic illustration of another configuration of the intervertebral-disc-treatment system of FIG. 1, in accordance with an application of the present invention;
- (23) FIG. 22 is a schematic illustration of an eye-treatment system for treating an eye of a subject, in accordance with an application of the present invention; and
- (24) FIG. 23 is a schematic illustration of a diabetic-retinopathy-treatment system for treating diabetic retinopathy, in accordance with an application of the present invention.

DETAILED DESCRIPTION OF APPLICATIONS

- (25) FIG. 1 is a schematic illustration of an intervertebral-disc-treatment system 20 for treating an intervertebral disc 30 of a subject, in accordance with an application of the present invention. Intervertebral-disc-treatment system 20 comprises (a) an electrode 22, which is configured to be implanted (typically chronically) partially within a nucleus pulposus 42 of disc 30, and (b) one or more extra-pulposus exposed electrode surfaces 44, which are configured to be implanted (typically chronically) outside nucleus pulposus 42, in electrical communication with disc 30, in a vicinity of an external surface of an annulus fibrosus 46 of disc 30, e.g., in physical contact with the external surface or not in physical contact with the external surface. Intervertebral-disc-treatment system 20 further comprises implantable (typically chronically implantable) or external control circuitry 50, which is typically electrically coupled, by one or more electrode leads 52, to (a) exposed electrode surfaces 44 and (b) electrode 22, such as via pin 76, described hereinbelow with reference to FIG. 2.
- (26) Typically, a healthcare worker, such as a physician, activates control circuitry 50 to provide the functions described herein. Activating the control circuitry may include configuring parameters and/or functions of the control circuitry (such as using a separate programmer or external controller), or activating the control circuitry to perform functions pre-programmed in the control circuitry. Control circuitry 50 typically comprises appropriate memory, processor(s), and hardware running software that is configured to provide the functionality of the control circuitry described herein.
- (27) Reference is made to FIG. 2, which is a schematic illustration of electrode 22, in accordance

with an application of the present invention. Electrode **22** comprises a wire **60**, which typically has a wire diameter $D_{sub.W}$ of at least 50 microns (more typically at least 75 microns, such as at least 90 microns, e.g., at least 95 microns), no more than 150 microns (more typically no more than 125 microns, such as no more than 110 microns, e.g., no more than 105 microns), and/or between 50 microns (more typically 75 microns, such as 90 microns, e.g., 95 microns) and 150 microns (more typically 125 microns, such as 110 microns, e.g., 105 microns), such as 100 microns.

(28) Wire **60** includes: a non-electrically-insulated current-application longitudinal segment **62**, which, in the absence of any applied forces, is helically coiled (referred to simply as “coiled” herein, including in the claims) and typically has (a) a mean pitch $P_{sub.1}$ of at least 1.1 (e.g., at least 1.2, such as at least 1.4), no more than 2 (e.g., no more than 1.8, such as no more than 1.6), and/or between 1.1 and 2 (e.g., between 1.2 and 1.8, such as between 1.4 and 1.6, e.g., 1.5) times the wire diameter $D_{sub.W}$, and (b) an entire longitudinal length $L_{sub.1}$ of at least 5 mm (e.g., at least 10 mm), no more than 35 mm (e.g., no more than 25 mm), and/or between 5 and 35 mm (e.g., between 10 and 25 mm), and an electrically-insulated lead longitudinal segment **64**, which typically has an entire longitudinal length $L_{sub.2}$ of at least 5 mm (e.g., at least 10 mm), no more than 80 mm (e.g., no more than 50 mm), and/or between 5 and 80 mm (e.g., between 10 and 50 mm), in the absence of any applied forces.

(29) As used in the present application, including in the claims, the pitch of a coil is the distance from any point on the coil to the corresponding point on an adjacent coil measured parallel to the central longitudinal axis. As used in the present application, including in the claims, the longitudinal length of segment is measured in a line along a central longitudinal axis of the segment (rather than being measured around the coil itself of the wire).

(30) Typically, current-application longitudinal segment **62** extends to a distal end **66** of wire **60**. For some applications, electrode **22** has a total length of at 50 mm (e.g., at least 70 mm), no more than 110 mm (e.g., no more than 90 mm), and/or between 50 and 110 mm (e.g., between 70 and 90 mm), such as 80 mm.

(31) For some applications, electrically-insulated lead longitudinal segment **64** is coated with parylene, or another biocompatible non-conducting material known in the art.

(32) For some applications, current-application longitudinal segment **62** has an outer coil diameter $C_{sub.1}$ of at least 3 (e.g., at least 4), no more than 7 (e.g., no more than 6), and/or between 3 and 7 (e.g., between 4 and 6), e.g., 5, times the wire diameter $D_{sub.W}$, in the absence of any applied forces. Alternatively or additionally, for some applications, the outer coil diameter $C_{sub.1}$ is at least 400 microns (e.g., at least 450 microns), no more than 600 microns (e.g., no more than 550 microns), and/or between 400 and 600 microns (e.g., between 450 and 550 microns), e.g., 500 microns, in the absence of any applied forces.

(33) Typically, because of the above-mentioned dimensions, including the pitch, current-application longitudinal segment **62** is highly flexible (both in its ability to bend and its ability to longitudinally stretch and contract), which reduces the application of forces on intervertebral disc **30**, and thus generally causes no or less trauma to intervertebral disc **30**, including nucleus pulposus **42**, than a more rigid electrode might cause, particularly over time during repeated motion of the disc.

(34) For some applications, lead longitudinal segment **64** is coiled, in the absence of any applied forces, such as shown in the figures. For some applications, lead longitudinal segment **64** has one or more of the following dimensions, in the absence of any applied forces: a mean pitch $P_{sub.2}$ that is greater than (such as at least 125% of, e.g., at least 150% of, at least 175% of, at least 200% of, or at least 250% of) the mean pitch $P_{sub.1}$ of current-application longitudinal segment **62**, a mean pitch $P_{sub.2}$ of at least 2 (e.g., at least 2.2, such as at least 2.4), no more than 3 (e.g., no more than 2.8, such as no more than 2.6), and/or between 2 and 3 (e.g., between 2.2 and 2.8, such as between 2.4 and 2.6), e.g., 2.5, times the wire diameter $D_{sub.W}$, an outer coil diameter $C_{sub.2}$ that equals between 90% and 110% (e.g., 100%) of the outer coil diameter $C_{sub.1}$ of current-application longitudinal segment **62**, and/or an outer coil diameter $C_{sub.2}$ of at least 3 (e.g., at least 4), no

more than 7 (e.g., no more than 6), and/or between 3 and 7 (e.g., between 4 and 6), e.g., 5, times the wire diameter $D_{sub.W}$.

(35) Typically, providing the above-mentioned pitches enables the thorough application of insulation to electrically-insulated lead longitudinal segment **64**, particularly if a vapor deposition process is used to apply the insulation (e.g., parylene). In addition, lead longitudinal segment **64** may have a greater pitch than current-application longitudinal segment **62** in part because lead longitudinal segment **64** generally does not require as much flexibility as current-application longitudinal segment **62**.

(36) For other applications, lead longitudinal segment **64** is not coiled, in the absence of any applied forces; for example, lead longitudinal segment **64** may be straight or zigzag-shaped (configurations not shown).

(37) For some applications, wire **60** further includes an intermediate longitudinal segment **70**, which (a) is longitudinally between current-application longitudinal segment **62** and lead longitudinal segment **64**, (b) typically has an entire longitudinal length $L_{sub.3}$ of at least 1 mm (e.g., at least 2 mm), no more than 6 mm (e.g., no more than 4 mm), and/or between 1 and 6 mm (e.g., between 2 and 4 mm), in the absence of any applied forces, and (c) in the absence of any applied forces, either (i) is coiled, and has a mean pitch $P_{sub.3}$ greater than outer coil diameter $C_{sub.1}$ of current-application longitudinal segment **62**, as shown in FIG. 2, or (ii) is not coiled, as shown in FIGS. 3A-B. (As used in the present application, including the claims, when a segment is described as “longitudinally between” two other segments, the segment is not necessarily adjacent to the two other segments. In other words, additional segments, which may or may not be mentioned, may longitudinally intervene between the segment and one or both of the two other segments.)

(38) For some applications, intermediate longitudinal segment **70** is electrically insulated along at least a longitudinal portion of intermediate longitudinal segment **70**.

(39) For some applications in which intermediate longitudinal segment **70** is coiled, intermediate longitudinal segment **70** has one or more of the following dimensions, in the absence of any applied forces: a mean pitch $P_{sub.3}$ of at least 125%, no more than 250%, and/or between 125% and 250% of the outer coil diameter $C_{sub.1}$ of current-application longitudinal segment **62**, a mean pitch $P_{sub.3}$ of at least 5 (e.g., at least 6, such as at least 9), no more than 20 (e.g., no more than 13, such as no more than 11), and/or between 5 and 20 (e.g., between 6 and 13, such as between 9 and 11), e.g., 10, times the wire diameter $D_{sub.W}$, a mean pitch $P_{sub.3}$ that equals (a) at least mean pitch $P_{sub.1}$ of current-application longitudinal segment **62**, and (b) no more than mean pitch $P_{sub.2}$ of lead longitudinal segment **64**, an outer coil diameter $C_{sub.3}$ equal to between 90% and 110% (e.g., 100%) of the outer coil diameter $C_{sub.1}$ of current-application longitudinal segment **62**, and/or an outer coil diameter $C_{sub.3}$ of at least 3 (e.g., at least 4), no more than 7 (e.g., no more than 6), and/or between 3 and 7 (e.g., between 4 and 6) times the wire diameter $D_{sub.W}$.

(40) Reference is now made to FIG. 3A, which is a schematic illustration of another configuration of electrode **22**, in accordance with an application of the present invention. This configuration is the same as the configuration described hereinabove with reference to FIG. 2, except that intermediate longitudinal segment **70** is not coiled, in the absence of any applied forces.

(41) For some applications, wire **60** further includes a partially non-electrically-insulated pin-connector longitudinal segment **72**. Lead longitudinal segment **64** is longitudinally between current-application longitudinal segment **62** and pin-connector longitudinal segment **72**. For these applications, intervertebral-disc-treatment system **20** typically comprises an electrode assembly **74**, which comprises electrode **22** and a pin **76**, which is mechanically fixed to and in electrical communication with pin-connector longitudinal segment **72**. Pin **76** and pin-connector longitudinal segment **72** are permanently connected, typically by welding (e.g., laser or spot welding).

(42) For some applications, such as shown, pin-connector longitudinal segment **72** is coaxial with a central longitudinal axis **80** of electrically-insulated lead longitudinal segment **64**, in the absence of

any applied forces.

(43) For some applications, such as shown in FIGS. 2 and 3A, pin-connector longitudinal segment 72 is coiled, in the absence of any applied forces. For some of these applications, such as shown in FIGS. 2 and 3A, coiled pin-connector longitudinal segment 72 is welded to the outside of pin 76.

(44) For some applications, pin-connector longitudinal segment 72 has one or more of the following dimensions, in the absence of any applied forces: a mean pitch $P_{sub.4}$ that equals at least 125%, no more than 250%, and/or between 125% and 250% of the outer coil diameter $C_{sub.2}$ of current-application longitudinal segment 62, a mean pitch $P_{sub.4}$ of at least 5, no more than 20 (e.g., no more than 10, such as no more than 7), and/or between 5 and 20 (e.g., between 5 and 10, such as between 3 and 7 times), e.g., 7, times the wire diameter $D_{sub.W}$, an outer coil diameter $C_{sub.4}$ that equals between 90% and 110% (e.g., 100%) of the outer coil diameter $C_{sub.2}$ of current-application longitudinal segment 62, an outer coil diameter $C_{sub.4}$ of at least 3 (e.g., at least 4), no more than 7 (e.g., no more than 6), and/or between 3 and 7 (e.g., between 4 and 6), e.g., 5, times the wire diameter $D_{sub.W}$, and/or an entire longitudinal length $L_{sub.4}$ of between 1 and 4 mm.

(45) Reference is now made to FIG. 3B, which is a schematic illustration of yet another configuration of electrode 22, in accordance with an application of the present invention.

(46) In this configuration, pin 76 is shaped as a tube 77 that defines a lumen. An end portion 78 of pin-connector longitudinal segment 72 may be a straight section located along the axis of pin-connector longitudinal segment 72. For example, the connection may be fixated by crimping the tube of pin 76 on the wire of pin-connector longitudinal segment 72. The coating (e.g., parylene coating) covers this section as well. Typically, the only part of pin 76 that is not be coated is the proximal end thereof, which serves as an electrical connector pin (by being insertable into a female electrical connector, typically after the electrode has been implanted). The electrical path is thus electrically connected to the control circuitry 50 and insulated from the body. For some applications, a seal 82 (e.g., comprising silicone) is provided that surrounds the connector/pin interface with electrode lead 52 to control circuitry 50. Optionally, seal 82 is shaped so as to define wings 84, which may be used to anchor the system.

(47) For some applications, one or more of current-application longitudinal segment 62, lead longitudinal segment 64, intermediate longitudinal segment 70, and pin-connector longitudinal segment 72 has an inner coil diameter of at least 0.25 (e.g., at least 0.27), no more than 0.36 (e.g., no more than 0.33), and/or between 0.25 and 0.36 (e.g., between 0.27 and 0.33), e.g., 0.6, times the respective outer coil diameter of the segment, in the absence of any applied forces.

(48) Typically, wire 60 has the same diameter $D_{sub.W}$ along all of current-application longitudinal segment 62, lead longitudinal segment 64, intermediate longitudinal segment 70, and pin-connector longitudinal segment 72. Alternatively, diameter $D_{sub.W}$ varies between one or more of current-application longitudinal segment 62, lead longitudinal segment 64, intermediate longitudinal segment 70, and/or pin-connector longitudinal segment 72.

(49) For some applications, the coil of one or more longitudinal segments of electrode 22 is filled with a pharmaceutical, such as an antibiotic pharmaceutical, an anti-inflammation pharmaceutical, or an analgesic pharmaceutical.

(50) Reference is now made to FIG. 4, which is a schematic illustration of a 3-mm long portion 90 of current-application longitudinal segment 62, in accordance with an application of the present invention. This configuration is applicable to the configurations of electrode 22 of FIGS. 2, 3, and/or 5. For some applications, when a central longitudinal axis 92 of 3-mm long portion 90 (measured along central longitudinal axis 92 when the portion is straight, in the absence of any applied forces) of current-application longitudinal segment 62 is curved in a plane only until all adjacent turns 94 of coiled wire 60 of portion 90 touch one another, central longitudinal axis 92 of the portion has a radius of curvature $R_{sub.C}$ of at least 5 mm, no more than 10 mm, and/or between 5 and 10 mm.

(51) Such a radius of curvature provides current-application longitudinal segment **62** with the flexibility to curve and flex. If, on the one hand, adjacent turns **94** of coiled wire **60** instead touched one other in the absence of any applied forces, current-application longitudinal segment **62** would not have such flexibility to curve. If, on the other hand, mean pitch $P_{sub.1}$ of current-application longitudinal segment **62** were substantially larger, current-application longitudinal segment **62** might be too stiff.

(52) For some applications, lead longitudinal segment **64** has the curvature parameters described hereinabove with reference to FIG. 4.

(53) Reference is now made to FIG. 5, which is a schematic illustration of another configuration of electrode **22**, in accordance with an application of the present invention. In this configuration, electrode **22** further comprises a water-permeable element **96** that runs along at least a longitudinal portion **98** of intermediate longitudinal segment **70**, thereby enhancing fluid flow through annulus fibrosus **46** caused by the application of the voltage, to a level greater than possible through other portions of annulus fibrosus **46**. For some applications, water-permeable element **96** comprises a membrane. Alternatively or additionally, for some applications, water-permeable element **96** comprises an acid selected from the group consisting of: glycolic acid and lactic acid. For some applications, water-permeable element **96** is pre-attached to longitudinal portion **98**. Alternatively, water-permeable element **96** may be placed separately into the bore through annulus fibrosus **46**.

(54) For some of these applications, intermediate longitudinal segment **70** is coiled, in the absence of any applied forces, such as described hereinabove with reference to FIG. 2, and water-permeable element **96** is disposed within the channel defined by the coil. Alternatively or additionally, intermediate longitudinal segment **70** is partially or entirely embedded in intermediate longitudinal segment **70**.

(55) Reference is again made to FIG. 1. At least a portion of electrode **22** is implanted in a body of a subject, typically such that: current-application longitudinal segment **62** is disposed at least partially in intervertebral disc **30** of the subject, typically entirely in intervertebral disc **30** (either entirely in nucleus pulposus **42** of intervertebral disc **30**, or partially in a nucleus pulposus of intervertebral disc **30** and partially in annulus fibrosus **46** of intervertebral disc **30**, as shown in FIG. 1), at least a portion of lead longitudinal segment **64** is disposed in the body of the subject outside intervertebral disc **30**, and if provided, intermediate longitudinal segment **70** is disposed at least in part in annulus fibrosus **46** of intervertebral disc **30**; providing the relatively large pitch for intermediate longitudinal segment **70**, or not coiling intermediate longitudinal segment **70** (as described hereinabove with reference to FIG. 3), may reduce the likelihood that the portion of intermediate longitudinal segment **70** that is within annulus fibrosus **46** forms a tunnel, which might cause leakage of fluid from nucleus pulposus **42**.

(56) Reference is now made to FIG. 6, which is a flow chart illustrating a method **100** of manufacturing electrode **22**, in accordance with an application of the present invention, and to FIG. 7, which is a schematic illustration of a polymer boot **102** used in manufacturing method **100**, in accordance with an application of the present invention. Manufacturing method **100** may be used to manufacture the configurations of electrode **22** of FIGS. 2, 3, and/or 5. Manufacturing method **100** begins at an electrode provision step **110** by providing electrode **22**, in any of the configurations described hereinabove that include intermediate longitudinal segment **70**, before electrically-insulated lead longitudinal segment **64** has been coated with an electrically-insulating coating. As mentioned above, intermediate longitudinal segment **70**, in the absence of any applied forces, either (a) is coiled and has a mean pitch $P_{sub.3}$ that is greater than an outer coil diameter $C_{sub.1}$ of current-application longitudinal segment **62** or (b) is not coiled.

(57) At a polymer boot provision step **112**, polymer boot **102** is provided; for some applications, polymer boot **102** comprises a silicone boot. Polymer boot **102** is shaped so as to define an enclosed space **116** with an opening **118** having a perimeter.

(58) At masking step **120**, a portion **126** of electrode **22** is masked by placing boot **102** on portion

126 of electrode **22**, such that: current-application longitudinal segment **62** is within enclosed space **116**, lead longitudinal segment **64** is outside boot **102**, and intermediate longitudinal segment **70** passes through opening **118** such that the perimeter of opening **118** forms a tight seal with at least a portion of intermediate longitudinal segment **70**.

(59) At a vapor deposition chamber step **130**, boot **102** and electrode **22** are placed into a vapor deposition chamber **134** (which is illustrated highly schematically in FIG. 7). At a parylene coating step **140**, a parylene coating (or another biocompatible non-conducting material known in the art) is applied to lead longitudinal segment **64** by vapor deposition within deposition chamber **134**, as is known in the vapor deposition art. Boot **102** and electrode **22** are removed from vapor deposition chamber **134**, and boot **102** is removed from electrode **22**.

(60) This masking technique, in combination with the geometry of electrode **22**, solves the problem of how to tightly mask current-application longitudinal segment **62** during vapor deposition of electrically-insulating parylene onto lead longitudinal segment **64**. The perimeter of opening **118** of boot **102**, when applied to the high-coil-pitch, or coil-lacking, intermediate longitudinal segment **70**, tightly squeezes the intermediate longitudinal segment, thereby preventing gaseous leakage of the parylene into the boot and onto current-application longitudinal segment **62**. Such tight squeezing would not be possible if intermediate longitudinal segment **70** were more tightly coiled (i.e., had a lower coil pitch).

(61) Reference is now made to FIG. 8, which is schematic illustrations of an implantation system **200** for implanting at least a portion of electrode **22** in a body of a subject, in accordance with an application of the present invention. Implantation system **200** may be used to implant the configurations of electrode **22** of FIGS. 2, 3, and/or 5. Implantation system **200** comprises the following components, each of which is described in detail hereinbelow: a hollow insertion needle **210**, which, at a proximal end thereof, is axially fixed to a needle-connection fitting **212**; insertion needle **210** is used for inserting electrode **22** partially into intervertebral disc **30**, as described hereinbelow with reference to FIGS. 11A-B; typically, insertion needle **210** has a length of between 60 and 120 mm, such as between 80 and 100 mm, e.g., 90 mm; a spacer **216**, which defines a channel therethrough for insertion of insertion needle **210** from a proximal side **218** of spacer **216** to a distal side **220** of spacer **216**, until needle-connection fitting **212** abuts proximal side **218** of spacer **216**; spacer **216** serves to limit a depth of penetration of insertion needle **210**, as described hereinbelow with reference to FIGS. 11A-B; an electrode loader **224**, which is used to advance electrode **22** into insertion needle **210**, as described hereinbelow with reference to FIGS. 13A-15B; and a needle-withdrawal handle **228**, which is used to withdrawn insertion needle **210** from disc **30** after delivery of electrode **22** into the disc, as described hereinbelow with reference to FIGS. 16A-18C.

(62) For some applications, one or more (e.g., all) of the components of implantation system **200** are provided as a kit, typically in sterile packaging **202**. The components of implantation system **200** may be provided in various stages of assembly, depending on packaging needs and other factors. Typically, electrode **22** is also provided in the kit, such as in the same sterile packaging **202** as the components of implantation system **200**.

(63) Reference is now made to FIGS. 9A-B, which are schematic illustrations of several components of implantation system **200**, in accordance with an application of the present invention. FIG. 9A shows needle-withdrawal handle **228** attached to spacer **216**, while insertion needle **210** passes through the channel of spacer **216**. FIG. 9A also shows needle-connection fitting **212** fixed to insertion needle **210** and disposed within needle-withdrawal handle **228**. FIG. 9B shows the relationship of the above-mentioned components.

(64) Reference is now made to FIG. 10, which is a flow chart illustrating a method **240** of implanting at least a portion of electrode **22** in a body of a subject, in accordance with an application of the present invention, and FIGS. 11A-20, which are schematic illustrations of steps of implantation method **240**, in accordance with an application of the present invention.

Implantation method **240** may be used to implant the configurations of electrode **22** of FIGS. **2**, **3**, and/or **5**. Implantation method **240** uses implantation system **200**, described hereinabove with reference to FIGS. **8** and **9A-B**.

(65) FIGS. **11A-B** are schematic illustrations of a needle insertion step **242** of implantation method **240**, in accordance with an application of the present invention. Before inserting insertion needle **210** into intervertebral disc **30**, insertion needle **210** (which is axially fixed to a needle-connection fitting **212**) is inserted through spacer **216**. For example, needle-connection fitting **212** may be female, in which case needle-connection fitting **212** typically comprises a female Luer taper. Spacer **216** serves to limit a depth penetration of insertion needle **210**, so that the insertion needle does not penetrate the far side of annulus fibrosus **46** of disc **30**. Optionally, a length of spacer **216** is adjustable, or a set of spacers **216** having different respective lengths is provided, in order to accommodate discs **30** of different sizes.

(66) Also before inserting insertion needle **210** into intervertebral disc **30**, an anti-coring stylet **244** is inserted through needle-connection fitting **212** and spacer **216** and into insertion needle **210**, with the distal end of anti-coring stylet **244** advanced to the distal end of insertion needle **210**. For some applications, anti-coring stylet **244** has a beveled distal end **246** that conforms to the beveled distal end of insertion needle **210**.

(67) At needle insertion step **242**, insertion needle **210** (with anti-coring stylet **244** therein) is partially inserted into disc **30**, until spacer **216** contacts an external surface of annulus fibrosus **46**. After insertion, the distal end of insertion needle **210** (and the distal end of anti-coring stylet **244**) typically is near, but does not penetrate, the far side of annulus fibrosus **46**. Alternatively, the distal end of insertion needle **210** (and the distal end of anti-coring stylet **244**) is inserted into the far side of annulus fibrosus **46**, e.g., to ultimately allow electrode **22** serve as an anchor. For some applications, the angle of penetration of the insertion needle is about 30 degrees.

(68) FIG. **12** is a schematic illustration of an anti-coring stylet **244** retraction step **248** of implantation method **240**, in accordance with an application of the present invention. At anti-coring stylet **244** retraction step **248**, anti-coring stylet **244** is retracted, leaving insertion needle **210** in disc **30**, and spacer **216** contacting the external surface of annulus fibrosus **46**.

(69) FIGS. **13A-C** are schematic illustrations of an electrode loader alignment step **252** of implantation method **240**, in accordance with an application of the present invention. Electrode loader **224** comprises an electrode-loader connection fitting **254**, which, for example, may be male, in which case electrode-loader connection fitting **254** typically comprises a male Luer taper, which is mounted near a distal end of a hollow loader needle **256**. Typically, loader needle **256** has a length of between 60 and 120 mm, such as between 80 and 100 mm, e.g., 90 mm.

(70) Electrode **22** is loaded in loader needle **256**, with a distal end of electrode **22** disposed near a distal end of loader needle **256**. Electrode loader **224** also comprises a loader stylet **258**, which is loaded partially within loader needle **256** and disposed such that a distal end of loader stylet **258** abuts a proximal end of electrode **22** (typically a proximal end of pin **76** of electrode **22**). Typically, loader stylet **258** is not beveled at its distal end (i.e., it typically has a straight distal end). Loader stylet **258** is held in place by a connecting element **260**, which may comprise a Luer connector, such that the distal end of loader stylet **258** can push electrode **22** distally, as described hereinbelow with reference to FIGS. **15A-B**. At electrode loader alignment step **252**, electrode loader **224** is aligned with insertion needle **210**.

(71) FIGS. **14A-B** are schematic illustrations of an electrode-loader connection step **264** of implantation method **240**, in accordance with an application of the present invention. At electrode-loader connection step **264**, electrode loader **224** is connected to insertion needle **210**, such as by connecting electrode-loader connection fitting **254** to needle-connection fitting **212**, such as, for example, by way of rotation. As a result of this connection, a distal end of loader needle **256** abuts a proximal end of insertion needle **210**.

(72) FIGS. **15A-B** are schematic illustrations of an electrode advancement step **268** of implantation

method **240**, in accordance with an application of the present invention. At electrode advancement step **268**, loader stylet **258** is gently advanced distally within loader needle **256** and pushes electrode **22** distally from loader needle **256** into insertion needle **210**, which, as mentioned above, is disposed partially in disc **30**. For example, electrode **22** may penetrate about 2 cm into disc **30**. Electrode-loader connection fitting **254** is disconnected from needle-connection fitting **212**, at an electrode-loader disconnection step **272** (not shown), and electrode loader **224** is removed.

(73) FIGS. **16A-C** are schematic illustrations of a needle-withdrawal-handle alignment step **276** of implantation method **240**, in accordance with an application of the present invention. Needle-withdrawal handle **228** comprises a handle stylet **278**, which is typically axially fixed within the needle-withdrawal handle. At needle-withdrawal-handle alignment step **276**, needle-withdrawal handle **228** and handle stylet **278** are aligned with insertion needle **210** (and spacer **216**).

(74) FIGS. **17A-B** are schematic illustrations of a needle-withdrawal-handle connection step **282** of implantation method **240**, in accordance with an application of the present invention. At needle-withdrawal-handle connection step **282**, needle-withdrawal handle **228** is connected to insertion needle **210**, such as by connecting needle-withdrawal handle **228** to spacer **216** (such as by friction, snapping on, and/or rotation). Upon the connection being made, needle-connection fitting **212** is disposed within needle-withdrawal handle **228**, near a distal end of the needle-withdrawal handle, and the distal end of handle stylet **278** is disposed within needle-connection fitting **212** abutting the proximal end of electrode **22** (typically the proximal end of pin **76** of electrode **22**).

(75) FIGS. **18A-C** are schematic illustrations of an insertion-needle withdrawal step **286** of implantation method **240**, in accordance with an application of the present invention. At insertion-needle withdrawal step **286**, needle-connection fitting **212** is withdrawn proximally within needle-withdrawal handle **228**, while needle-withdrawal handle **228** is held stationary. As mentioned above with reference to FIGS. **11A-B**, insertion needle **210** is axially fixed to needle-connection fitting **212**. As a result, needle-connection fitting **212**, as it is proximally withdrawn, withdraws insertion needle **210** from disc **30** (insertion needle **210** is slidable with respect to spacer **216**). Handle stylet **278**, which is axially fixed with respect to needle-withdrawal handle **228** and abuts the proximal end of electrode **22**, prevents proximal motion of electrode **22**. As a result, electrode **22** remains generally axially stationary, partially inserted in disc **30**.

(76) FIG. **19** is a schematic illustration of a needle-withdrawal-handle withdrawal step **290** of implantation method **240**, in accordance with an application of the present invention. At needle-withdrawal-handle withdrawal step **290**, needle-withdrawal handle **228**, with spacer **216** still attached thereto, is gently withdrawn proximally, thereby releasing electrode **22** and leaving electrode **22** implanted partially in disc **30** and partially in the body of the subject outside the disc.

(77) FIG. **20** is a schematic illustration of electrode **22** implanted partially in disc **30** and partially in the body of the subject outside the disc, in accordance with an application of the present invention. For example, about 2 cm of electrode **22** may be disposed in disc **30**, and the remainder (e.g., about 5 to 6 cm) may be disposed outside the disc.

(78) Reference is again made to FIG. **1**. In some applications of the present invention, intervertebral-disc-treatment system **20** comprises: at least one intra-pulposus exposed electrode surface **280** (which is electrically conductive), which is configured to be implanted in nucleus pulposus **42** of intervertebral disc **30**; a plurality of extra-pulposus exposed electrode surfaces **44** (e.g., at least 3, no more than 10, and/or between 3 and 10, such as exactly 3), which are configured to be implanted outside nucleus pulposus **42**, in electrical communication with intervertebral disc **30**; and control circuitry **50**, which is (a) electrically coupled to the at least one intra-pulposus exposed electrode surface **280** and the plurality of extra-pulposus exposed electrode surfaces **44**, (b) configured to separately control at least two of the plurality of extra-pulposus exposed electrode surfaces **44**; for example, control circuitry **50** may be electrically coupled to the extra-pulposus exposed electrode surfaces separately via separate electrical conductors.

(79) Providing the plurality of separately-controllable extra-pulposus exposed electrode surfaces **44**

distributes the generation of hydrogen, thereby reducing any local build-up of hydrogen at any single electrode surface.

(80) For some applications, control circuitry **50** is configured to: repeatedly assume a pressure-increasing mode of operation in alternation with an oxygen-generating mode of operation, in the pressure-increasing mode of operation, (a) configure the at least one intra-pulposus exposed electrode surface **280** to be a cathode, and one or more of the plurality of extra-pulposus exposed electrode surfaces **44** to be one or more respective anodes, and (b) electroosmotically drive fluid into nucleus pulposus **42** to increase pressure in intervertebral disc **30**, by applying a first mean voltage of less than 1.23 V (sometimes known in the art as the “electrolysis voltage”) (e.g., less than 1 V, such as less than 500 mV, e.g., less than 300 mV) between the at least one intra-pulposus exposed electrode surface **280** and the one or more of the plurality of extra-pulposus exposed electrode surfaces **44**, and in the oxygen-generating mode of operation, (a) configure the at least one intra-pulposus exposed electrode surface **280** to be an anode, and the plurality of extra-pulposus exposed electrode surfaces **44** to be a respective plurality of cathodes, and (b) generate oxygen within nucleus pulposus **42** by electrolysis, by applying a second mean voltage of at least 1.23 V (e.g., at least 1.5 V, such as at least 2 V) between the at least one intra-pulposus exposed electrode surface **280** and the plurality of extra-pulposus exposed electrode surfaces **44**.

(81) The increase in fluid in nucleus pulposus **42** during the pressure-increasing mode of operation generally treats or prevents further degeneration of the disc caused at least in part by loss of fluid. The applied current may also help introduce nutritional substances into the disc. The generation of oxygen within nucleus pulposus **42** during the oxygen-generating mode generally treats hypoxia, which, if untreated, sometimes causes disc degeneration. The generation of oxygen may also improve glucose metabolism, while reducing lactic acid generation.

(82) For some applications, control circuitry **50** is configured to apply direct current, e.g., with an average amplitude of between 1 and 5 mA. For some applications, the control unit is configured to apply the direct current as a series of pulses. For some applications, the control unit is configured to apply the direct current as the series of pulses with a duty cycle of between 1% and 50%.

(83) Typically, control circuitry **50** is not configured to actively balance the applied positive and negative charges. Rather, control circuitry **50** is configured to allow the passive balancing of the applied positive and negative charges.

(84) Typically, control circuitry **50** is configured to separately control all of the plurality of extra-pulposus exposed electrode surfaces **44**; for example, control circuitry **50** may be electrically coupled to the extra-pulposus exposed electrode surfaces separately via separate electrical conductors.

(85) For some applications, as shown in FIG. 1, the at least one intra-pulposus exposed non-electrically-insulated electrode surface **280** comprises current-application longitudinal segment **62** of electrode **22**, described hereinabove. Alternatively, the at least one intra-pulposus exposed electrode surface **280** does not comprise non-electrically-insulated current-application longitudinal segment **62** of electrode **22**; instead, another electrode is provided; for example, electrodes may be used that are described in U.S. Pat. No. 8,577,469 and/or U.S. application Ser. No. 14/982,187, filed Dec. 29, 2015, which published as US Patent Application Publication 2017/0182317. both of which are assigned to the assignee of the present application and incorporated herein by reference, and are optionally implanted during a conventional surgical procedure to repair disc **30** and/or nucleus pulposus **42**, including a standard approach for inserting a needle in disc **30**. Alternatively, an electrode known in the art is used.

(86) For some applications, control circuitry **50** is configured, during a period of time, to assume (a) the pressure-increasing mode of operation at least 10 times for an aggregate first duration and (b) the oxygen-generating mode of operation at least 10 times for an aggregate second duration that is less than 10% (e.g., less than 5%, such as less than 1%) of the aggregate first duration. By way of example and not limitation, control circuitry **50** may assume the oxygen-generating mode of

operation for 1-30 seconds (e.g., 1-30 seconds every hour), and assume the pressure-increasing mode of operation at other times. Although control circuitry **50**, when in the oxygen-generating mode of operation, may electroosmotically drive fluid out of nucleus pulposus **42** and thus decrease pressure in intervertebral disc **30**, because the aggregate second duration is so much less than the aggregate first duration, the aggregate effect of the application of voltages is an increase in pressure in intervertebral disc **30**.

(87) For some applications, control circuitry **50** is configured to, in the oxygen-generating mode of operation, generate oxygen within nucleus pulposus **42** by electrolysis, by applying the second mean voltage between the at least one intra-pulposus exposed electrode surface **280** and respective different subsets of the plurality of extra-pulposus exposed electrode surfaces **44** at respective different times. For some applications, each of the subsets consists of exactly one of the plurality of extra-pulposus exposed electrode surfaces **44**. Activating extra-pulposus exposed electrode surfaces **44** at different times further distributes the generation of hydrogen, thereby further reducing any local build-up of hydrogen at any single electrode surface.

(88) For some applications, intervertebral-disc-treatment system **20** further comprises a sensor, which is configured to sense a parameter indicative of a quantity of the oxygen generated by the electrolysis. Control circuitry **50** is configured to modulate the oxygen-generating mode of operation responsively to the sensed parameter. For some applications, control circuitry **50** is configured to modulate the oxygen-generating mode of operation by modulating a duration of one or more occurrences of the oxygen-generating mode of operation. Alternatively or additionally, control circuitry **50** is configured to modulate the oxygen-generating mode of operation by modulating an electrical parameter of the second mean voltage (e.g., the voltage, amplitude, duty cycle, and/or frequency). For some applications, the sensed parameter is an oxygen concentration in nucleus pulposus **42** and/or a pH in nucleus pulposus **42**. The sensor may be provided in the application of the present invention described immediately hereinbelow with reference to FIGS. **1** and **21**.

(89) Reference is still made to FIG. **1**, and is additionally made to FIG. **21**, which is a schematic illustration of another configuration of intervertebral-disc-treatment system **20**, in accordance with an application of the present invention. In some applications of the present invention, intervertebral-disc-treatment system **20** comprises: at least one intra-pulposus exposed electrode surface **280**, which is configured to be implanted in nucleus pulposus **42** of intervertebral disc **30**; one or more extra-pulposus exposed electrode surfaces **44**, which (a) are configured to be implanted outside nucleus pulposus **42**, in electrical communication with intervertebral disc **30**, and (b) have an aggregate electrically-exposed surface area of at least 3 cm², such as at least 4 cm², e.g., at least 5 cm²; and control circuitry **50**, which is electrically coupled to the at least one intra-pulposus exposed electrode surface **280** and one or more extra-pulposus exposed electrode surfaces **44**.

(90) For some applications, control circuitry **50** is configured to: repeatedly assume a pressure-increasing mode of operation in alternation with an oxygen-generating mode of operation, in the pressure-increasing mode of operation, (a) configure the at least one intra-pulposus exposed electrode surface **280** to be a cathode, and the one or more extra-pulposus exposed electrode surfaces **44** to be one or more respective anodes, and (b) electroosmotically drive fluid into nucleus pulposus **42** to increase pressure in intervertebral disc **30**, by applying a first mean voltage of less than 1.23 V (e.g., less than 1 V, such as less than 500 mV, e.g., less than 300 mV) between the at least one intra-pulposus exposed electrode surface **280** and the one or more extra-pulposus exposed electrode surfaces **44**, and in the oxygen-generating mode of operation, (a) configure the at least one intra-pulposus exposed electrode surface **280** to be an anode, and the one or more extra-pulposus exposed electrode surfaces **44** to be a respective plurality of cathodes, and (b) generate oxygen within nucleus pulposus **42** by electrolysis, by applying a second mean voltage of at least 1.23 V (e.g., at least 2 V) between the at least one intra-pulposus exposed electrode surface **280** and

the one or more extra-pulposus exposed electrode surfaces **44**.

(91) The increase in fluid in nucleus pulposus **42** during the pressure-increasing mode of operation generally treats or prevents further degeneration of the disc caused at least in part by loss of fluid. The applied current may also help introduce nutritional substances into the disc. The generation of oxygen within nucleus pulposus **42** during the oxygen-generating mode generally treats hypoxia, which, if untreated, sometimes causes disc degeneration. The generation of oxygen may also improve glucose metabolism, while reducing lactic acid generation. Providing the relatively large aggregate electrically-exposed surface area of at least 3 cm² distributes the generation of hydrogen, thereby reducing any local build-up of hydrogen at the electrode-tissue interface.

(92) For some applications, such as shown in FIG. **21**, intervertebral-disc-treatment system **20** comprises exactly one extra-pulposus exposed electrode surface **44** having an electrically-exposed surface area of at least 3 cm². Alternatively, an external surface of a can of control circuitry **50** serves as extra-pulposus exposed electrode surface **44** (configuration not shown). For other applications, such as shown in FIG. **1**, intervertebral-disc-treatment system **20** comprises a plurality of extra-pulposus exposed electrode surfaces **44**.

(93) For some applications, control circuitry **50** is configured to apply direct current, e.g., with an average amplitude of between 1 and 5 mA. For some applications, the control unit is configured to apply the direct current as a series of pulses. For some applications, the control unit is configured to apply the direct current as the series of pulses with a duty cycle of between 1% and 50%.

(94) For some applications, control circuitry **50** is configured, during a period of time, to assume (a) the pressure-increasing mode of operation at least 10 times for an aggregate first duration and (b) the oxygen-generating mode of operation at least 10 times for an aggregate second duration that is less than 10% (e.g., less than 5%, such as less than 1%) of the aggregate first duration. By way of example and not limitation, control circuitry **50** may assume the oxygen-generating mode of operation for a few seconds every hour, and assume the pressure-increasing mode of operation at other times. Although control circuitry **50**, when in the oxygen-generating mode of operation, may electroosmotically drive fluid out of nucleus pulposus **42** and thus decrease pressure in intervertebral disc **30**, because the aggregate second duration is so much less than the aggregate first duration, the aggregate effect of the application of voltages is an increase in pressure in intervertebral disc **30**.

(95) Reference is again made to FIGS. **1** and **21**. For some applications, control circuitry **50** is configured to drive intra-pulposus exposed electrode surface **280** and the one or more extra-pulposus exposed electrode surfaces **44** to electroosmotically drive fluid between inside and outside nucleus pulposus **42** based on a circadian cycle of the subject. For some applications, a housing containing control circuitry **50** is injectable, with an anchor at the proximal end. One or more extra-pulposus exposed electrode surfaces **44** are fixed to an external surface of the housing. For example, the housing may be implanted immediately posterior to the spinal column. For some applications, control circuitry **50** is configured to be implanted subcutaneously, if the housing containing the control circuitry is small. Alternatively, for some applications, control circuitry **50** is configured to be implanted or elsewhere in the subject's body, if the housing of the control circuitry is larger (e.g., includes batteries).

(96) For some applications, control circuitry **50** is driven by an external controller that is in wireless or wired communication with control circuitry **50**. For some applications, the external controller is mounted on a bed of the subject (e.g., disposed within a mattress), and is configured to activate control circuitry **50** only at night, and/or only when the subject is sleeping. Such nighttime activation may coincide with and support the filling phase of the disc, and thus be therapeutic even though the patient experiences more pain during the day. Alternatively or additionally, control circuitry **50** is activated during the daytime, i.e., over the course of the day, because the pressure is higher in the disc during application of vertical and mechanical load on the disc, which causes the disc to lose fluid; the activation may thus reduce maximum damage to the disc. Further

alternatively, the control circuitry is activated generally constantly or regularly intermittently (e.g., one hour on/one hour off). For some applications, control circuitry **50** is activated during rest of the subject, rather than during activity; for example, an accelerometer may be provided to identify movement vs. rest of the subject.

(97) For some applications, control circuitry **50** is configured to provide the subject with control of activation of control circuitry **50**, e.g., in response to activity or pain. For example, the control may be provided from the subject's telephone (e.g., smartphone) or other electronic device.

(98) For some applications, the method further comprises replacing nucleus pulposus **42** with an artificial substitute material before implanting intra-pulposus exposed electrode surface **280**.

(99) A first experiment was conducted on behalf of the inventors to study the feasibility of using some of the techniques described hereinabove with reference to FIG. **1** to hydrate and dehydrate a spinal disc, and the impact of the voltage application on the disc mass. The experiment evaluated three electrical protocols: (a) negative voltage inside the nucleus pulposus of the disc vs. outside the disc, (b) positive voltage inside the nucleus pulposus of the disc vs. outside the disc, and (c) control (no voltage applied to the nucleus pulposus of the disc). It was found that application of a negative voltage inside the nucleus pulposus of the disc enhanced the hydration of the disc, as compared to a positive voltage or no voltage. No dehydration effect was observed with application of a positive voltage to the nucleus pulposus.

(100) The experiment used a total of six fresh bovine tail discs. In order to achieve equilibrium, the discs were placed in a saline solution for a period of one hour prior to application of the voltages. The discs were then weighed. The discs were randomly assigned to the experimental groups as follows: (a) two specimens—negative voltage inside the nucleus pulposus of the disc vs. outside the disc, (b) two specimens—positive voltage inside the nucleus pulposus of the disc vs. outside the disc, and (c) two specimens—control (no voltage applied to the nucleus pulposus of the disc).

(101) The discs were placed inside a vessel and fully submerged in saline solution. One electrode was inserted in the approximate center of the nucleus pulposus of each of the discs in experimental groups (a) and (b). The electrode was electrically-insulated except at its tip, and was designed to allow submersion in liquid. The electrode was inserted laterally (i.e., through the annulus of the disc). A second, ring electrode was placed within the saline solution surrounding the disc.

(102) Voltages of (a) -1 V and (b) $+1$ V were applied between the electrodes in the two experimental groups (a) and (b), respectively. These voltages were selected to be lower than the electrolysis voltage of water of about 1.2 V. After a period of two hours, the discs were removed and weighed again.

(103) As set forth in Table 1 below, all of the discs increased in mass during the voltage-application period. The mass of the discs to which the negative internal voltage was applied increased by 4.7% and 5.5%, while the mass of the other discs (positive internal voltage and control) increased by between 2.0% and 2.6%.

(104) The inventors hypothesize that all of the discs absorbed liquid, while the application of the negative internal voltage contributed to an additional absorption of 2-3%. The application of the positive internal voltage did not result in dehydration of the disc.

(105) TABLE-US-00001

	TABLE 1	Mass after 1	Mass after 2	Mass	Mass	Disc	hour	immersion																											
Voltage	hours	voltage	change	change	# [g]	(internal)	application	[g] [g] [%]																											
1	3.342	-1 V	3.527	0.185	5.5%	2	4.384	-1 V	4.590	0.206	4.7%	3	6.552	$+1$ V	6.720	0.168	2.6%	4	4.558	$+1$ V	4.651	0.093	2.0%	5	7.346	0	7.531	0.185	2.5%	6	6.074	0	6.209	0.135	2.2%

(106) A second experiment was conducted on behalf of the inventors to study the feasibility of using some of the techniques described hereinabove with reference to FIG. **1** to hydrate a spinal disc and the impact of the voltage application on the disc mass. The experiment evaluated two electrical protocols: (a) negative voltage inside the nucleus pulposus of the disc vs. outside the disc, and (b) control (no voltage applied to the nucleus pulposus of the disc). It was found that application of a negative voltage inside the nucleus pulposus of the disc enhanced the hydration of

the disc, as compared to no voltage. Higher voltage markedly increased the mass gain.

(107) The experiment used a total of six fresh bovine tail discs. The discs were randomly assigned to the experimental groups as follows: (a) three specimens—negative voltage inside the nucleus pulposus of the disc vs. outside the disc, and (b) three specimens—control (no voltage applied to the nucleus pulposus of the disc).

(108) The discs were weighed, and then placed inside a vessel and fully submerged in saline solution. One electrode was inserted in the approximate center of the nucleus pulposus of each of the discs in the experimental group (a). The electrode was electrically-insulated except at its tip, and was designed to allow submersion in liquid. The electrode was inserted laterally (i.e., through the annulus of the disc). A second, ring electrode was placed within the saline solution surrounding the disc.

(109) A voltage of -1 V was applied between the electrodes in the experimental group (a). One hour after the beginning of application of the voltage, a first pair of two of the discs (one negative voltage, one control) were removed and weighed. In the two remaining negative voltage discs, the voltage was increased to -3 V.

(110) Two hours after the beginning of application of the voltage, a second pair of two of the discs (one negative voltage, one control) were removed and weighed.

(111) Three hours after the beginning of application of the voltage, a third pair of two of the discs (one negative voltage, one control) were removed and weighed.

(112) As set forth in Table 2 below, all of the discs increased in mass during the voltage-application period. In each pair, the disc to which the voltage was applied increased in mass more than the control disc did. Increasing the voltage from -1 V to -3 V resulted in a markedly increased mass gain. It was noted, however, that the -3 V voltage application resulted in electrolysis of the solution, which was expected since the electrolysis threshold of water is about 1.2 V. This electrolysis was observed as bubbles and discoloration in the solution.

(113) The inventors hypothesize that all of the discs absorbed liquid, while the application of the negative internal voltage contributed to an additional absorption.

(114) TABLE-US-00002

Starting mass [g]	Duration [h]	Ending mass [g]	mass change [%]
6.047	1	6.345	4.93%
6.227	0	6.420	3.10%
5.988	3	6.420	3.10%
5.988	3	6.420	3.10%
6.605	2	6.605	10.30%
5.192	0	5.192	0
5.444	2	5.444	4.85%
4.484	1	4.484	0
5.262	3	5.262	17.35%
4.236	0	4.236	0
4.619	3	4.619	9.04%

(115) The inventors hypothesize that application of -3 V, although possibly not suitable for clinical use, served as a proxy for the effectiveness of longer-term voltage application at a lower voltage, such as -1 V.

(116) As mentioned above, the discs were placed in a saline-dye solution during the experiment. The dye was methylene blue. After weighing the discs, the discs were also dissected and inspected for dye penetration. In general, dye penetration was not observed in the discs.

(117) Reference is now made to FIG. 22, which is a schematic illustration of an eye-treatment system 300 for treating an eye 310 of a subject, in accordance with an application of the present invention. Eye-treatment system 300 is typically used for treating glaucoma. Eye-treatment system 300 comprises: a first exposed electrode surface 320, which is configured to be implanted in a vitreous cavity 322 of eye 310; a second exposed electrode surface 324, which is configured to be implanted in a body of the subject at a site 326 outside vitreous cavity 322; and control circuitry 350, which is typically electrically coupled, by one or more electrode leads 352, to first and second exposed electrode surfaces 320 and 324.

(118) For some applications, control circuitry 350 is configured to: repeatedly assume a pressure-decreasing mode of operation in alternation with an oxygen-generating mode of operation, in both the pressure-decreasing mode of operation and the oxygen-generating mode of operation, configure first exposed electrode surface 320 to be an anode, and second exposed electrode surface 324 to be a cathode, in the pressure-decreasing mode of operation, electroosmotically drive fluid from

vitreous cavity **322** to outside vitreous cavity **322** to decrease pressure in vitreous cavity **322**, by applying a first mean voltage of less than 1.23 V (e.g., less than 1 V, such as less than 500 mV, e.g., less than 300 mV) between first and second exposed electrode surfaces **320** and **324**, and in the oxygen-generating mode of operation, generate oxygen within vitreous cavity **322** by electrolysis, by applying a second mean voltage of at least 1.23 V (e.g., at least 2 V) between first and second exposed electrode surfaces **320** and **324**.

(119) The decrease in fluid in vitreous cavity **322** during the pressure-decreasing mode of operation generally treats glaucoma by reducing intraocular pressure. The generation of oxygen within vitreous cavity **322** during the oxygen-generating mode treats diabetic retinopathy, in which insufficient oxygen is naturally provided to the retina.

(120) For some applications, site **326** is within a Schlemm's canal **354** of the subject, and second exposed electrode surface **324** is configured to be implanted within Schlemm's canal **354**. In these applications, control circuitry **50** is configured to, in the pressure-decreasing mode of operation, electroosmotically drive the fluid from vitreous cavity **322** to Schlemm's canal **354** to decrease the pressure in vitreous cavity **322**, by applying the first mean voltage between first and second exposed electrode surfaces **320** and **324**.

(121) For some applications, control circuitry **50** is configured, during a period of time, to assume (a) the pressure-decreasing mode of operation at least 10 times for an aggregate first duration and (b) the oxygen-generating mode of operation at least 10 times for an aggregate second duration that is less than 10% (e.g., less than 5%, such as less than 1%) of the aggregate first duration.

(122) For some applications, control circuitry **50** is configured to apply direct current, e.g., with an average amplitude of between 1 and 5 mA. For some applications, the control unit is configured to apply the direct current as a series of pulses. For some applications, the control unit is configured to apply the direct current as the series of pulses with a duty cycle of between 1% and 50%.

(123) For some applications, eye-treatment system **300** further comprises a sensor, which is configured to sense a parameter indicative of a quantity of the oxygen generated by the electrolysis. Control circuitry **50** is configured to modulate the oxygen-generating mode of operation responsively to the sensed parameter. For some applications, control circuitry **50** is configured to modulate the oxygen-generating mode of operation by modulating a duration of one or more occurrences of the oxygen-generating mode of operation. Alternatively or additionally, control circuitry **50** is configured to modulate the oxygen-generating mode of operation by modulating an electrical parameter of the second mean voltage (e.g., the voltage, amplitude, duty cycle, and/or frequency). For some applications, the sensed parameter is an oxygen concentration in vitreous cavity **322** and/or a pH in vitreous cavity **322**.

(124) For some applications, control circuitry **50** is configured to detect a pressure difference between vitreous cavity **322** and outside vitreous cavity **322**, and modulate the pressure-decreasing mode of operation responsively to the pressure difference parameter in response to the detected pressure difference.

(125) Reference is now made to FIG. **23**, which is a schematic illustration of a diabetic-retinopathy-treatment system **400** for treating diabetic retinopathy, in accordance with an application of the present invention. Diabetic-retinopathy-treatment system **400** comprises control circuitry **420** and one or more electrodes **430** (e.g., at least 2, no more than 20, and/or between 2 and 20). The one or more electrodes **430** are configured to be implanted penetrating a retina **440** of eye **310** (as shown), or placed against an external surface of retina **440** (i.e., the surface facing into the interior of the eye) (configuration not shown).

(126) Control circuitry **420** is configured to generate oxygen within retina **440** by electrolysis, by (a) configuring the one or more electrodes **430** to be anodes, and (b) applying a mean voltage of at least 1.23 V (e.g., at least 1.5 V, such as at least 2 V) between the one or more electrodes **430** and one or more return cathodes **444** (e.g., an electrically-conductive casing of control circuitry **420**). The generated oxygen treats diabetic retinopathy, which is characterized by damage to blood

vessels in the retina, which reduces the oxygen supply to the retina.

(127) For some applications, diabetic-retinopathy-treatment system **400** comprises an antenna **446**, which is configured to wirelessly receive energy for powering control circuitry **420**. For example, antenna **446** may have an annular shape and be mounted around a lens **448** of the eye, either the natural lens or an intraocular lens (IOL) that is implanted to serve as a support for antenna **446**. Alternatively, antenna **446** has a different structure and/or is implanted at a different site in the eye or near the eye. Antenna **446** may be connected to control circuitry **420** by one or more conductive wires **449**.

(128) Typically, electrodes **430** are implanted near a macula **442** of retina **440**, most typically not on macula **442** itself, in order to avoid interfering with the patient's vision. For example, as shown, diabetic-retinopathy-treatment system **400** may comprise a ring **450** that is sized to be placed around macula **442**, and electrodes **430** are fixed at different respective sites along ring **450** around macula **442**. Optionally, control circuitry **420** is also disposed along, or slightly outside, ring **450**.

(129) In some applications of the present invention, the techniques and apparatus described herein are combined with techniques and apparatus described in one or more of the following applications, which are assigned to the assignee of the present application and are incorporated herein by reference: U.S. Pat. No. 8,577,469 to Gross; US Patent Application Publication 2014/0324128 to Gross; and U.S. patent application Ser. No. 14/982,187, filed Dec. 29, 2015, which published as US Patent Application Publication 2017/0182317.

(130) It will be appreciated by persons skilled in the art that the present invention is not limited to what has been particularly shown and described hereinabove. Rather, the scope of the present invention includes both combinations and subcombinations of the various features described hereinabove, as well as variations and modifications thereof that are not in the prior art, which would occur to persons skilled in the art upon reading the foregoing description.

Claims

1. A method comprising: providing an electrode, which includes a wire that (a) has a wire diameter of between 75 and 125 microns and (b) includes: (1) a non-electrically-insulated current-application longitudinal segment, which, in the absence of any applied forces, is coiled and has (i) an outer coil diameter of between 3 and 7 times the wire diameter, and (ii) an entire longitudinal length of between 5 and 35 mm, and (2) an electrically-insulated lead longitudinal segment, which has an entire longitudinal length of at least 10 mm, in the absence of any applied forces; and implanting at least a portion of the electrode in a body of a subject.

2. The method according to claim 1, wherein the current-application longitudinal segment has an outer coil diameter of between 3 and 7 times the wire diameter, in the absence of any applied forces.

3. The method according to claim 2, wherein the outer coil diameter of the current-application longitudinal segment is between 4 and 6 times the wire diameter, in the absence of any applied forces.

4. The method according to claim 1, wherein the current-application longitudinal segment has an outer coil diameter of between 400 and 600 microns, in the absence of any applied forces.

5. The method according to claim 1, wherein the current-application longitudinal segment has a mean pitch of between 1.1 and 2 times the wire diameter.

6. The method according to claim 5, wherein the mean pitch of the current-application longitudinal segment is between 1.2 and 1.8 times the wire diameter, in the absence of any applied forces.

7. The method according to claim 1, wherein the wire diameter is between 90 and 110 microns.

8. The method according to claim 1, wherein the entire longitudinal length of the current-application longitudinal segment is between 10 and 25 mm, in the absence of any applied forces.

9. The method according to claim 1, wherein the current-application longitudinal segment extends

to a distal end of the wire.

10. The method according to claim 1, wherein the lead longitudinal segment is coiled, in the absence of any applied forces.

11. The method according to claim 10, wherein the lead longitudinal segment has a mean pitch that is greater than the mean pitch of the current-application longitudinal segment, in the absence of any applied forces.

12. The method according to claim 11, wherein the mean pitch of the lead longitudinal segment is at least 125% of the mean pitch of the current-application longitudinal segment, in the absence of any applied forces.

13. The method according to claim 10, wherein the lead longitudinal segment has a mean pitch of between 2 and 3 times the wire diameter, in the absence of any applied forces.

14. The method according to claim 10, wherein the lead longitudinal segment has an outer coil diameter that equals between 90% and 110% of an outer coil diameter of the current-application longitudinal segment, in the absence of any applied forces.

15. The method according to claim 10, wherein the lead longitudinal segment has an outer coil diameter of between 3 and 7 times the wire diameter, in the absence of any applied forces.

16. The method according to claim 1, further comprising activating control circuitry, which is electrically coupled to the electrode, to drive the electrode to apply a current.

17. The method according to claim 1, wherein the non-electrically-insulated current-application longitudinal segment, in the absence of any applied forces, is shaped so as to define an empty inner channel therethrough.

18. The method according to claim 1, wherein implanting the at least a portion of the electrode includes implanting the at least a portion of the electrode such that: the current-application longitudinal segment is disposed at least partially in an intervertebral disc of the subject, and at least a portion of the lead longitudinal segment is disposed in the body of the subject outside the intervertebral disc.

19. The method according to claim 18, wherein implanting the at least a portion of the electrode includes implanting the at least a portion of the electrode such that the current-application longitudinal segment is disposed entirely in a nucleus pulposus of the intervertebral disc.

20. The method according to claim 18, wherein the wire further includes an intermediate longitudinal segment, which (a) is longitudinally between the current-application longitudinal segment and the lead longitudinal segment, (b) has an entire longitudinal length of between 1 and 6 mm, in the absence of any applied forces, and (c) in the absence of any applied forces, either (i) is coiled, and has a mean pitch greater than an outer coil diameter of the current-application longitudinal segment or (ii) is not coiled, and wherein implanting the at least a portion of the electrode includes implanting the at least a portion of the electrode such that the intermediate longitudinal segment is disposed at least in part in an annulus fibrosus of the intervertebral disc.

21. The method according to claim 20, wherein the intermediate longitudinal segment is electrically insulated along at least a longitudinal portion of the intermediate longitudinal segment.

22. The method according to claim 20, wherein, in the absence of any applied forces, the intermediate longitudinal segment is coiled, and has the mean pitch greater than the outer coil diameter of the current-application longitudinal segment.
